

Phase transition of WTAP regulates m⁶A modification of interferon-stimulated genes

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This **important** study demonstrates that interferon beta stimulation induces WTAP transition from aggregates to liquid droplets, coordinating m⁶A modification of a subset of mRNAs that encode interferon-stimulated genes and restricting their expression. The evidence presented is **solid**, supported by microscopy, immunoprecipitations, m⁶A sequencing, and ChIP, to show that WTAP phosphorylation controls phase transition and its interaction with STAT1 and the methyltransferase complex.

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Abstract

N⁶-methyladenosine (m⁶A) is the most prevalent modification of mRNA which controls diverse physiological processes. Although m⁶A modification has been reported to regulate type I interferon (IFN) responses by targeting the mRNA of IFN- β and the interferon stimulated genes (ISGs), the detailed mechanism of how m⁶A methyltransferase complex (MTC) rapidly responds to conduct the modification on nascent mRNA during IFN- β stimulation remains largely unclear. Here, we demonstrate that WTAP, the adaptor protein of m⁶A MTC, undergoes dephosphorylation-regulated phase transition from aggregates to liquid-like condensates under IFN- β stimulation, thereby mediates m⁶A modification of a subset of ISG mRNAs to restrict their expression. The phase transition of WTAP promotes the interaction with nucleus-translocated transcription factor STAT1, which recruits MTC to the promoter regions of ISGs and directing the co-transcriptional m⁶A modification on ISG mRNAs. Collectively, our findings reveal a novel regulatory role of WTAP phase transition in manipulating signaling pathways and fine-tuning immune response by orchestrating dynamic m⁶A modification through the cooperation of transcription factors and MTC.

Introduction

Innate immunity is the first defense line against invading pathogens, mainly through the induction of type I interferon (IFN) and IFN-stimulated genes (ISGs) (Goubau, Deddouch, & Reis e Sousa, 2013). Once the type I IFN is stimulated and initiated, the signal should be tightly restricted to maintain the homeostasis of immune responses (Henault et al., 2016; Schneider, Chevillotte, & Rice, 2014). In the past few years, a large number of studies have reported the post-translational modification of signaling proteins that regulate type I IFN signaling, such as phosphorylation (Uddin et al., 2002; Wen Z, Zhong Z, & Jr., 1995), ubiquitination (Cui et al., 2014; Qin et al., 2016) and methylation (Mowen et al., 2001). However, whether post-transcriptional modification of mRNA affects type I IFN responses remains largely unclear.

Methylation at the N^6 position of adenosine (m^6A) is the most pervasive post-transcriptional modification of mRNA. Deposition of m^6A is catalyzed by methyltransferase complex (MTC), including a key adaptor protein Wilm's tumor-associated protein (WTAP) and key catalyze proteins methyltransferase-like 3 (METTL3) and methyltransferase-like 14 (METTL14) (Schapira, 2016). The m^6A -modified RNAs are recognized by YT521B homology (YTH) domain family proteins, including YTHDC1, YTHDC2, YTHDF1, YTHDF2 and YTHDF3. m^6A modification have been found to regulate different cellular processes (Knuckles & Bühler, 2018; Meyer & Jaffrey, 2017), such as differentiation of stem cells (Geula et al., 2015), circadian clock (Zhong et al., 2018), splicing (Z. Zhang et al., 2010), translation (X. Wang et al., 2015) and destabilization of mRNA (X. Wang et al., 2014). Meanwhile, m^6A modification can be reversed by demethylase fat mass and obesity-associated gene (FTO) and AlkB homolog 5 (ALKBH5) (Jia et al., 2011; Zheng et al., 2013). m^6A modification is critical for newborn mRNA produced by specific stimulation to participate in further physiological activities. While m^6A modification of *IFN-β* mRNA suppresses antiviral responses during viral infection (Rubio, Depledge, Bianco, Thompson, & Mohr, 2018; Winkler et al., 2019), this modification also paradoxically enhances antiviral response through different regulation of ISG transcripts, such as enhancing translation of IFIT1 (McFadden et al., 2021) and prolonging *STAT1* and *IRF1* mRNAs stability (L. Wang et al., 2020), demonstrating the complexity of m^6A in immune modulation. Given the crucial role of precise and timely m^6A modification on mRNA in fine-tuning antiviral responses, further investigation is required to clarify the function of non-catalytic protein WTAP and how MTC responds to IFN-β stimulation.

Here, we observed the phase transition from gel-like aggregates to liquid-like condensates of WTAP driven by IFN-β-induced dephosphorylation. Based on the results from the WTAP mutation assay, we revealed that under IFN-β stimulation, the liquid-phase of WTAP cooperated with transcription factor STAT1 to recruit MTC at the promoter regions of ISG, mediating the m^6A modification on newborn ISG mRNAs and regulated the expression of a subset of ISGs, thereby modulating antiviral type I IFN responses. Our study sheds light on the mechanism by which phase transition of WTAP bridges transcription factors and MTC to fine-tune m^6A deposition on specific mRNAs.

Results

WTAP undergoes phase separation during IFN-β stimulation

To explore the detailed mechanism of WTAP-directed m^6A modification on ISG mRNAs, we first investigated the expression of WTAP during virus infection. We infected cells with RNA virus Vesicular Stomatitis Virus (VSV) or DNA virus Herpes Simplex Virus-1 (HSV-1), and conducted the immunofluorescence experiments. Surprisingly, we found WTAP was clustered and formed

condensate-like pattern in cells during virus infection (**Figure 1A** and Figure 1-figure supplement 1A). We then stimulated cells with IFN- β and observed the increasing number of WTAP condensates formation (**Figure 1B**). Previous studies have reported that some components in the methyltransferase complex such as METTL3 or YTHDFs undergo liquid-liquid phase separation (LLPS) during m⁶A modification process (Han et al., 2022; J. Li et al., 2022). Therefore, we suspected whether the IFN- β -mediated WTAP condensates indicated phase separation.

A considerable number of proteins undergo phase separation via interactions between intrinsically disordered regions (IDRs). IDRs contain more charged and polar amino acids to present multiple weakly interacting elements, and lack hydrophobic amino acids to show flexible conformations (Hou et al., 2024). By searching WTAP protein structure (within part of MTC complex) in protein structure bank (<https://www.rcsb.org/structure/7VF2>), we found that WTAP IDR was predicted to span amino acids from 245 to 396, giving a clue that WTAP might have the ability to undergo phase separation (**Figure 1C**). To directly prove the phase separation properties of WTAP *in vitro*, we purified the recombinant mCherry-WTAP protein, mixed recombinant mCherry-WTAP with physiological buffer and found that the buffer became turbid (**Figure 1D**). Observation under microscope showed that mCherry-WTAP automatically formed liquid-like condensates which could be reversed by 1,6-hexanediol (hex), an inhibitor of phase separation (**Figure 1E**). Phase diagram of WTAP phase separation was also established under different concentrations of potassium chloride and recombinant mCherry-WTAP (Figure 1-figure supplement 1B). Fusion of condensates were observed and larger condensates were formed *in vitro* (Figure 1-figure supplement 1C). These data together confirmed that WTAP underwent phase separation *in vitro*.

To further explore the protein properties, we detected the phase separation behavior of WTAP in cells. Gene expressing mCherry-WTAP was introduced in WTAP-KO HeLa cells to generate mCherry-WTAP-rescued HeLa cells, followed with IFN- β stimulation. A few WTAP condensates were discovered in untreated cells while an increasing number of WTAP condensates were observed under IFN- β stimulation. After hex treatment, WTAP condensates in IFN- β treated cells dissipated into a dispersed pattern (**Figure 1F**). Similarly, hex treatment in virus-infected cells also disrupted the formation of WTAP condensates (**Figure 1A** and Figure 1-figure supplement 1A). We next checked the mobility of WTAP condensates by fluorescence recovery after photobleaching (FRAP) experiments. Intriguingly, WTAP exhibited lower recovery intensity in untreated cells but higher mobility with rapid recovery intensity in IFN- β -treated cells (**Figure 1G**). These results suggested a potential dynamic transformation in WTAP phase separation. While WTAP underwent gel-like phase separation under basal conditions, IFN- β induced its phase transition to form the liquid-like condensates under physiological conditions.

Previous studies have uncovered that protein phase separation could be controlled by interaction between specific regions through multiple factors (Murthy et al., 2019; J. Wang et al., 2018), including electrostatic interactions (Boyko, Qi, Chen, Surewicz, & Surewicz, 2019), hydrophobic contacts (Reichheld, Muiznieks, Keeley, & Sharpe, 2017) or hydrogen bonds (Gabryelczyk et al., 2019). Therefore, we tried to figure out the phase separation-driven region of WTAP. We analyzed the net charge per region and hydropathy of WTAP, but no specific domains with opposite charge or high level of hydrophobicity were found (Figure 1-figure supplement 2A-B), indicating that electrostatic interaction and hydrophobic contacts might not be the key driving force for WTAP phase separation. We then analyzed the abundance of amino acids and found that Serine (Ser), Glutamine (Gln), Glutamate (Glu) and Leucine (Leu) were present at significantly higher levels than other amino acids within WTAP (**Figure 1C** and Figure 1-figure supplement 2C). Gln and Leu were mainly located in the N-terminal domain (NTD) while Ser was mostly enriched in the C-terminal domain (CTD), implying the different properties of NTD and CTD (**Figure 1C**). Notably, IDR of WTAP were predicted to cover the entire serine-rich CTD region. Previous researches reported that hydrogen bond between serine side chain mediated the

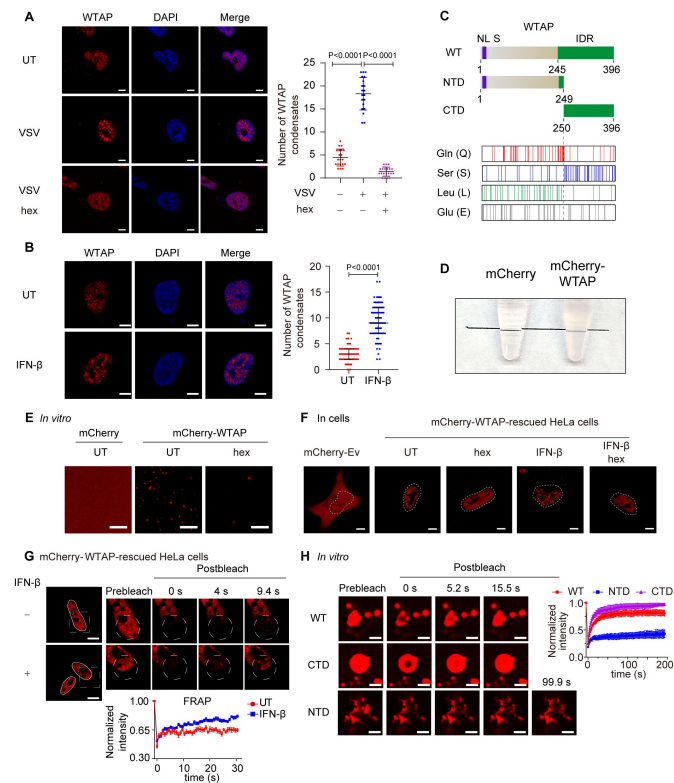


Figure 1

WTAP goes through phase transition with IFN-β stimulation.

(A) THP-1-derived macrophages were infected with Vesicular Stomatitis Virus (VSV) (m.o.i. = 0.1) for 24 hours together with or without 5% 1,6-hexanediol (hex) and 10 µg/mL digitonin for 2 hours or left untreated (UT). Endogenous WTAP was stained and imaged using confocal microscopy. The number of WTAP condensates that diameter over 0.4 µm of n = 20 cells were counted through ImageJ and shown. Scale bars indicated 5 µm.

(B) HeLa cells were placed on the dishes and stimulated with or without 10 ng/mL IFN-β for 1 hour at 37 °C. Endogenous WTAP was stained and imaged using confocal microscopy. The number of WTAP condensates that diameter over 0.4 µm of n = 80 cells were counted through ImageJ and shown. Scale bars indicated 5 µm.

(C) Domain structures (top) and distribution of amino acids (bottom) of WTAP protein. WT, wild type. NTD, N-terminal domain. CTD, C-terminal domain. NLS, nuclear localization signal. IDR, intrinsically disordered regions. Gln, Glutamine. Ser, Serine. Lue, Leucine. Glu, Glutamic acid.

(D) Tubes containing physiological buffer with recombinant mCherry (10 µM) or mCherry-WTAP (10 µM) were compared, in which recombinant mCherry-WTAP underwent phase separation.

(E) Foci formation of recombinant mCherry-WTAP (10 µM) with or without 5% 1,6-hexanediol (hex) *in vitro* was observed through confocal microscopy. Scale bars indicated 5 µm.

(F) Phase separation of mCherry-WTAP in mCherry-WTAP-rescued HeLa cells treated with or without 5% hex and 20 µg/mL digitonin were observed through confocal microscopy. Representative images of n = 20 cells were shown. Scale bars indicated 5 µm.

(G) mCherry-WTAP-rescued HeLa cells were placed on dishes and treated with or without 10 ng/mL IFN-β for 1 hour at 37 °C. After stimulation, bleaching of the WTAP foci was performed and quantification of fluorescence recovery after photobleaching (FRAP) of mCherry-WTAP aggregates was analyzed. The start time point of recovery after photobleaching was defined as 0 second. Representative images of n = 10 cells were shown. Scale bars indicated 5 µm.

(H) Recombinant mCherry-WT WTAP, NTD and CTD (10 µM) were mixed with the physiological buffer and placed on the dishes at 37°C (Prebleach). After incubation, bleaching was performed and quantification of FRAP of recombinant mCherry-WT WTAP, NTD and CTD were analyzed. Representative images of n = 6 condensates were shown, and the normalized intensity was measured and analyzed. The start time point of recovery after photobleaching was defined as 0 second. Scale bars indicated 2 µm.

All error bars, mean values ± SD, P-values were determined by unpaired two-tailed Student's *t*-test of n = 20 cells in (A) and n = 80 cells in (B). For (A-B and D-H), similar results were obtained for three independent biological experiments.

intermolecular interaction and LLPS of FUS. In addition, abnormal glutamine-repeat resulted in the formation of β -sheet and solid/gel-like aggregates with lower mobility (Perutz, 1996 [↗](#); Tanaka, Morishima, Akagi, Hashikawa, & Nukina, 2001 [↗](#)), which were related to neurodegenerative diseases like Huntington's disease (Gourfinkel-An et al., 1997 [↗](#)). Therefore, we wondered whether glutamine-rich NTD and serine-rich CTD of WTAP induced the distinct phase separation. As predicted, we found full-length WTAP and CTD WTAP formed liquid condensates, while NTD of WTAP clustered and formed aggregates with lower mobility, which was confirmed through FRAP experiments (**Figure 1H** [↗](#) and Figure 1-figure supplement 2D). Intriguingly, we further observed that full-length WTAP transitioned from liquid-like condensates to aggregates after incubation for several minutes while CTD of WTAP remained liquid phase (Figure 1-figure supplement 2E). Collectively, these data validated the different phase separation potential of NTD and CTD of WTAP.

IFN- β -mediated dephosphorylation induces the phase transition of WTAP

Since the serine-rich CTD played the key role in liquid condensates formation of WTAP, we wondered whether IFN- β -mediated WTAP phase transition through serine-phosphorylation by electrostatic repulsion among negatively charged phosphate. We first detected the phosphorylation level of WTAP through immunoprecipitation (IP) assay and found that phosphorylation of WTAP was significantly decreased under IFN- β stimulation (**Figure 2A** [↗](#)). Through IP assays, we aimed to identify the phosphatases that dephosphorylate WTAP, thus checked the interaction between WTAP and family of protein phosphatases (PPPs), and found that PPP4 presented the strongest interaction with WTAP (Figure 2-figure supplement 1A). IFN- β stimulation promoted the interaction between WTAP and PPP4 (**Figure 2B** [↗](#) and Figure 2-figure supplement 1B). Both knockdown of PPP4 and the potent PPP4 inhibitor fostriecin restrained the dephosphorylation of WTAP in IFN- β treated cells (**Figure 2C** [↗](#)). Potent PPP4 inhibitor fostriecin to detect, which significantly impaired the IFN- β -induced dephosphorylation of WTAP (Figure 2-figure supplement 1C). These data indicating that IFN- β regulated dephosphorylation of WTAP through PPP4.

To validate the effects of phosphorylation on the phase transition of WTAP, we first confirming the phosphorylation site of WTAP by mass spectrometry (MS) assays. MS results identified six phosphorylated serine/threonine sites within WTAP, while five of six phosphorylation sites were located in CTD of WTAP (**Figure 2D** [↗](#) and Figure 2-figure supplement 1D). We then constructed a series of phosphorylation mimic, serine/threonine to aspartate mutants (S/T-D) or phosphorylation-deficient, serine/threonine to alanine mutant (S/T-A) of WTAP, and observed the phase separation of WTAP mutants *in vitro*. We found that introducing three or more phosphorylation-mimic mutations in WTAP promoted protein aggregation (**Figure 2E** [↗](#) and Figure 2-figure supplement 1E). While the 5ST-D mutant induced aggregation, the 5ST-A mutant formed liquid condensates comparable to wild-type (WT) WTAP, which remained unphosphorylated in the *E. coli* expression system lacking endogenous kinase activity (**Figure 2F** [↗](#)). We next conducted FRAP experiments and found that mobility of WT WTAP as well as 5ST-A was significantly higher than that of its 5ST-D mutant *in vitro* (**Figure 2G** [↗](#)). However, hyperphosphorylated WT WTAP in untreated cells showed lower mobility as 5ST-D mutant compared to phosphorylation-deficient 5ST-A mutant, while IFN- β -induced dephosphorylation promoted the recovery efficiency of WT WTAP as higher as 5ST-A mutants of WTAP (**Figure 2H** [↗](#)). Additionally, we found the liquid condensates of WTAP after IFN- β stimulation was blunted in PPP4 knocked-down cells and fostriecin treatment (Figure 2-figure supplement 1F-H), revealed that IFN- β triggered PPP4-mediated dephosphorylation of WTAP controlled the phase transition of WTAP. Taken together, we validated hyperphosphorylation of WTAP tended to form aggregates with low mobility due to electrostatically repulsed between phosphorylated serine sites, while hypophosphorylation of WTAP promoted the mobility and forms liquid condensates. IFN- β -mediated dephosphorylation promotes the transition from aggregates to liquid-phase of WTAP both *in vitro* and in cells (**Figure 2I** [↗](#)).

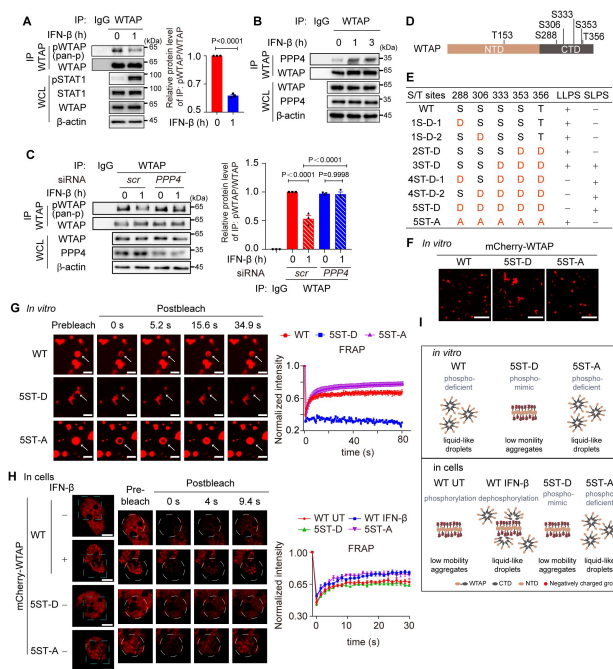


Figure 2.

IFN-β-mediated dephosphorylation of WTAP induces its phase transition.

(A) THP-1-derived macrophages were treated with 10 ng/mL IFN-β for 1 hour or left untreated. Whole cell lysate (WCL) was collected and immunoprecipitation (IP) experiment using anti-WTAP antibody or rabbit IgG was performed, followed with immunoblot. pWTAP was detected by anti-phosphoserine/threonine/tyrosine antibody (pan-p). Relative protein level was shown.

(B) THP-1-derived macrophages were treated with 10 ng/mL IFN-β for 1 hour or left untreated. WCL was collected and IP experiment using anti-WTAP antibody or rabbit IgG was performed, followed with immunoblot.

(C) THP-1-derived macrophages were transfected with *scramble* (scr) or *PPP4*-targeted siRNA and stimulated with or without 10 ng/mL IFN-β for 1 hour. WCL was collected and IP experiment with anti-WTAP antibody or rabbit IgG was performed, followed with immunoblot. pWTAP was detected by anti-phosphoserine/threonine/tyrosine antibody (pan-p). Relative protein level was measured and analyzed.

(D) mCherry-WTAP was purified from HEK 293T cells expressing mCherry-WTAP using anti-WTAP antibody and analyzed by mass spectrometry (MS) for the phosphorylation. Six phosphorylated residues were identified. Data of MS assay for five phosphorylated sites within CTD were shown in Figure 2-figure supplement 1D. (E-F) Recombinant mCherry-WTAP mutants with indicated serine/threonine (S/T) to aspartic acid (D) or alanine (A) mutation were listed in (E), while representative images of the phase-separated mCherry-WTAP WTAP, 5S-D and 5S-A mutants were shown in (F). Images of phase separation of other mCherry-WTAP mutants (10 μM) were shown in Figure 2-figure supplement 1E. Scale bars indicated 5 μm.

(G) Recombinant mCherry-WTAP WTAP (10 μM), 5S-D mutant (10 μM) or 5S-A mutant (10 μM) was mixed with the physiological buffer and placed on the dishes at 37°C (Prebleach). After incubation, bleaching was performed and quantification of fluorescence recovery after photobleaching (FRAP) of recombinant mCherry-WTAP WTAP, 5S-D or 5S-A mutant were analyzed. Representative images of n = 5 condensates were shown and the normalized intensity was measured and analyzed. The start time point of recovery after photobleaching was defined as 0 second. Arrow indicated the FRAP area while scale bars indicated 2 μm.

(H) mCherry-WTAP WTAP, 5S-D or 5S-A mutants-rescued HeLa cells were placed on the dishes and stimulated with or without 10 ng/mL IFN-β for 1 hour at 37°C. After seeding, bleaching of the WTAP foci was performed and quantification of FRAP of mCherry-WTAP aggregates was analyzed. Representative images of n = 10 cells were shown, and the normalized intensity was measured and analyzed. The start time point of recovery after photobleaching was defined as 0 second. Scale bars indicated 5 μm.

(I) Schematic figure of phase separation of WT WTAP, 5S-D and 5S-A in vitro (above) and WT WTAP with IFN-β stimulation or left untreated (UT), 5S-D and 5S-A in cells (below).

All error bars, mean values ± SD, P-values were determined by unpaired two-tailed Student's *t*-test of n = 3 independent biological experiments in (A and C). For (A-C and E-H), similar results were obtained for three independent biological experiments.

Phase transition of WTAP directs m⁶A modification on ISG mRNAs to regulate ISGs expression

Next, we aimed to determine how IFN- β -mediated WTAP phase transition functioned in ISG mRNAs. Previous reports have shown that m⁶A modification affected antiviral activity through ISG mRNAs stabilization or translation (McFadden et al., 2021 [\[1\]](#); L. Wang et al., 2020 [\[2\]](#)). To verify the function of WTAP on ISG mRNAs m⁶A modification, we generated *WTAP*^{sgRNA} THP-1 cells using the CRISPR-Cas9 system, and performed RNA sequencing (RNA-Seq) followed with bioinformatic analysis, and uncovered that 1,440 and 1,689 genes were up-regulated in response to IFN- β stimulation for 6 and 12 hours respectively (**Figure 3A-B** [\[3\]](#) and Figure 3-figure supplement 1A). Among this, 441 ISGs exhibited elevated expression in *WTAP*^{sgRNA} cells, including *IFIT1*, *IFIT2*, *OAS1* and *OAS2*, which were enriched in various immune-related pathways (**Figure 3C-D** [\[4\]](#) and Figure 3-figure supplement 1B). We also found that WTAP showed no effect on both the expression level and the phosphorylation level of transcription factors STAT1, STAT2 and IRF9 under IFN- β stimulation (Figure 3-figure supplement 1C-D), suggesting that changes of WTAP-regulated changes in ISGs expression were independent of phosphorylation level of indicated transcription factors.

We then collected the mRNA of control and *WTAP*^{sgRNA} cells stimulated with or without IFN- β to perform m⁶A methylated RNA immunoprecipitation followed by deep sequencing (MeRIP-Seq). Consensus m⁶A modified core motifs were enriched in our samples and the majority of WTAP-dependent m⁶A modification peaks were located in coding sequences (CDS) and 3'-untranslated regions (UTR) (**Figure 3E-F** [\[5\]](#) and Figure 3-figure supplement 1E-F). A group of ISGs that was m⁶A modified in control cells but showed diminished modification in *WTAP*^{sgRNA} cells were clustered in the IFN-associated pathway through MeRIP-seq assays, verifying the WTAP-dependent m⁶A modification of ISG mRNAs during IFN- β stimulation (Figure 3-figure supplement 1G). Notably, around 64.29% of core ISGs and 61.90% of WTAP down-regulated ISGs were m⁶A modified, which was consistent with the similar percentage in previous studies (McFadden et al., 2021 [\[1\]](#); Winkler et al., 2019 [\[6\]](#)) (**Figure 3G** [\[7\]](#) and Figure 3-figure supplement 1B). The topology of ISGs m⁶A sites was analyzed, revealing a strong preference for CDS localization compared to global m⁶A deposition (**Figure 3H** [\[8\]](#) and Figure 3-supplement 1H). MeRIP-quantitative real time-polymerase chain reaction (qPCR) assays confirmed the reduced m⁶A modification level on WTAP-down-regulated ISGs, including *IFIT1*, *IFIT2*, *OAS1* and *OAS2* (**Figure 3I** [\[9\]](#)). Given that m⁶A deposition was reported to affect various aspects of RNA, especially for the decay of mRNA (Oerum, Meynier, Catala, & Tisne, 2021 [\[10\]](#); Shi et al., 2017 [\[11\]](#); X. Wang et al., 2014 [\[12\]](#)), we also checked the mRNA stabilization by inhibiting synthesis of nascent mRNA using Actinomycin D (Act D), and the expression level of these ISGs by qPCR assay. Our results showed that WTAP deficiency stabilized ISG mRNAs, leading to their up-regulated expression (**Figure 3J-K** [\[13\]](#)).

To further delineate the function of WTAP phase separation on m⁶A modification, we treated cells with hex to disrupt the phase separation of WTAP, and detected the m⁶A deposition in cells with virus infection or IFN- β treatment. We found that virus infection or IFN- β stimulation induced m⁶A modified ISG mRNAs, and their m⁶A modification was disrupted by phase separation inhibitor hex (**Figure 4A** [\[14\]](#) and Figure 4-figure supplement 1A-B). Hex increased ISG mRNAs stability in control cells but had no effect in *WTAP*^{sgRNA} cells, suggesting WTAP phase separation was essential for ISG mRNAs m⁶A deposition and stabilization (Figure 4-figure supplement 1C). Next, we aimed to uncover the distinct ability to direct m⁶A modification of ISGs between aggregates or liquid-phase separated WTAP. Knockdown of *PPP4* or treatment with its inhibitor fostriecin, maintained WTAP in a hyperphosphorylated and aggregated status under IFN- β stimulation, and the m⁶A modification level on ISG mRNAs were surprisingly found to be decreased (**Figure 4B-C** [\[15\]](#)). Enhanced mRNA level was also detected with *PPP4* deficiency or inhibition (Figure 4-figure supplement 1D-E). We re-introduced WT WTAP, or its 5ST-D or 5ST-A mutant into *WTAP*^{sgRNA} THP-1 cells to generate WT WTAP, 5ST-D or 5ST-A THP-1 cells respectively (Figure 4-figure supplement 1F). Our results showed that m⁶A modification level of ISG mRNAs in

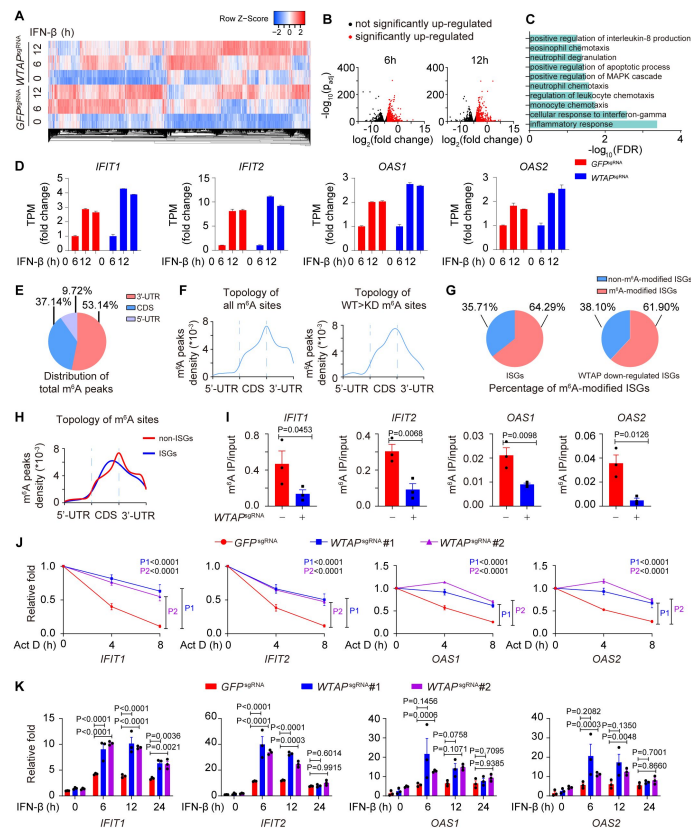


Figure 3.

WTAP is crucial for the m⁶A modification and expression of ISG mRNAs.

(A) Transcriptome sequencing analysis of control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages stimulated with 10 ng/mL IFN-β for 0, 6, 12 hours. The count-per-million (CPM) value of the ISGs was drawn by Heatmapper and clustered using Centroid Linkage approach.

(B) Volcano plots showing the changes in transcripts level of IFN-β up-regulated genes in *WTAP^{sgRNA}* THP-1-derived macrophages versus control (*GFP^{sgRNA}*) cells under IFN-β stimulation for 6 and 12 hours. Red dots indicated the significantly up-regulated genes in *WTAP^{sgRNA}* cells ($\log_2(\text{fold change}) > 1$ while adjusted P-value (p_{adj}) < 0.05).

(C) Gene ontology analysis for the WTAP down-regulated ISGs.

(D) TPMs showing the changes in transcripts level of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* in control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages stimulated with 10 ng/mL IFN-β for 6 or 12 hours or left untreated.

(E-F) Control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages were treated with 10 ng/mL IFN-β for 4 hours, m⁶A modification analyzed by m⁶A methylated RNA immunoprecipitation (IP) followed by deep sequencing (MeRIP-seq). Distribution (E) and topology (F) analysis (5'-untranslated regions (UTR), coding sequences (CDS) and 3'-UTR) of total m⁶A sites in control (*GFP^{sgRNA}*) cells or WTAP-dependent m⁶A sites.

(G) Percentage of m⁶A-modified ISGs including core ISGs or WTAP down-regulated ISGs were calculated and presented.

(H) Topology analysis (5'-UTR, CDS and 3'-UTR) of ISGs m⁶A sites and non-ISGs m⁶A sites.

(I) Control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages were treated with 10 ng/mL IFN-β for 4 hours. MeRIP-quantitative real time-polymerase chain reaction (qPCR) assay of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A-modified mRNA and input were shown (m⁶A IP/input).

(J) Control (*GFP^{sgRNA}*), *WTAP^{sgRNA}* #1 and *WTAP^{sgRNA}* #2 THP-1-derived macrophages were treated with 10 ng/mL IFN-β for 2 hours. After IFN-β treatment, medium with stimuli was replaced by 5 μM actinomycin D (Act D) for indicated time points. RNA was collected and detected by qPCR assay.

(K) Control (*GFP^{sgRNA}*), *WTAP^{sgRNA}* #1 and *WTAP^{sgRNA}* #2 THP-1-derived macrophages were treated with 10 ng/mL IFN-β for indicated time points. Expression of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* mRNA was analyzed through qPCR assays.

All error bars, mean values ± SEM, P-values were determined by unpaired two-tailed Student's *t*-test of *n* = 3 independent biological experiments in (I). All error bars, mean values ± SEM, P-values were determined by two-way ANOVA test of *n* = 3 independent biological experiments in (J-K).

WTAP^{sgRNA} and WTAP 5ST-D mutant cells was significantly attenuated than that of control cells, WT WTAP cells and WTAP 5ST-A mutant cells (**Figure 4D** [↗](#)). Hex suppressed m⁶A modification in WT WTAP and 5ST-A cells, but not in 5ST-D cells, implying the crucial role of liquid-phase separated WTAP in m⁶A deposition event (**Figure 4E** [↗](#)). Taken together, these results demonstrated that m⁶A modification of ISG mRNAs was mainly regulated by the liquid-phase separated WTAP dephosphorylated by PPP4 under IFN-β stimulation.

Liquid-phase separated WTAP recruits MTC and STAT1 on promoter regions to direct the m⁶A modification of ISG mRNAs

Although m⁶A modification of ISGs has been reported previously, the detailed mechanism of how WTAP directed m⁶A modification on ISG mRNAs and the distinct mechanism between WTAP aggregates and liquid-phase separated WTAP were still unknown. By analyzing expression levels of WTAP and other m⁶A methylation complex proteins, no significant alteration was observed during 6 hours of IFN-β stimulation (Figure 5-figure supplement 1A-B). By MeRIP-seq, we discovered the m⁶A modification sites of ISGs were preferentially localized to CDS region (**Figure 3H** [↗](#)), which aligned with co-transcriptional m⁶A patterns (Barbieri et al., 2017 [↗](#); H. Huang et al., 2019 [↗](#)). Additionally, previous researches have demonstrated the interaction between METTL3/WTAP and the transcription factor STAT5B and SMAD2/3 directed m⁶A modification to downstream genes, supporting transcription factors-mediated co-transcriptional m⁶A modification (Bertero et al., 2018 [↗](#); Bhattarai, Kim, Lim, & Choi, 2024 [↗](#)). Therefore, we wondered whether m⁶A modification of ISG mRNAs was directed co-transcriptionally by WTAP and IFN-β-activated transcription factors, including STAT1, STAT2 and IRF9 (Schneider et al., 2014 [↗](#)). Occupancy of the promoters of WTAP-regulated ISGs (2000 bp upstream of genes were analyzed, while 500 bp upstream of genes were considered as the proximal part of promoter) were predicted through directing using AnimalTFDB 3.0 website (<http://bioinfo.life.hust.edu.cn/AnimalTFDB/> [↗](#)) (Hu et al., 2019 [↗](#)). Most of the ISGs were occupied by STAT1 rather than STAT2 and IRF9 (**Figure 5A** [↗](#)), especially for the proximal part of promoter, pointing the possible role of STAT1 in WTAP-induced m⁶A modification. We then detected the interaction between WTAP, METTL3 and STAT1 through IP assay and found that the interaction between WTAP, METTL3 and nuclear-translocated STAT1 was enhanced under IFN-β stimulation (**Figure 5B** [↗](#), and Figure 5-figure supplement 1C). Additionally, the interaction between STAT1 and METTL3 was abolished in WTAP^{sgRNA} cells (**Figure 5C** [↗](#)). RIP-qPCR assays confirmed disrupted METTL3-ISG mRNAs interaction in WTAP^{sgRNA} cells, demonstrating the involvement and importance of WTAP in METTL3-dependent m⁶A deposition, and verifying the formation of STAT1-WTAP-METTL3-ISG mRNAs complex under IFN-β stimulation (**Figure 5D** [↗](#)).

To assess the contribution of phase-separated WTAP, we used hex to disrupt WTAP phase separation, and interaction with STAT1 were analyzed. The interaction between WTAP, METTL3 and STAT1 was dramatically inhibited with hex treatment both in cells and *in vitro* (**Figure 5E-F** [↗](#), and Figure 5-figure supplement 1D). By immunofluorescence (IF) experiments, we found IFN-β promoted the interaction between WTAP, METTL3 and nuclear-translocated STAT1 in cells, which could be abolished by hex (**Figure 5G** [↗](#)). The interaction between WTAP, METTL3 and STAT1 were enhanced in the liquid-phase mutant WTAP 5ST-A rescued cells, which could be impaired by hex treatment (Figure 5-figure supplement 1E). The condensation among recombinant GFP-STAT1, mCherry-WTAP and CFP-METTL3 *in vitro* were detected, and observed that WTAP was able to form condensates with STAT1 or METTL3 alone while condensation of STAT1 and METTL3 was significantly diminished in the absence of WTAP (**Figure 5H-I** [↗](#) and Figure 5-figure supplement 1F). Consistent with the results in cells, the interaction between WTAP and STAT1 was blunted by hex (Figure 5-figure supplement 1G). Notably, both 2% and 5% hex did not alter the PPP4c-mediated WTAP dephosphorylation level, confirming that hex inhibited WTAP phase separation

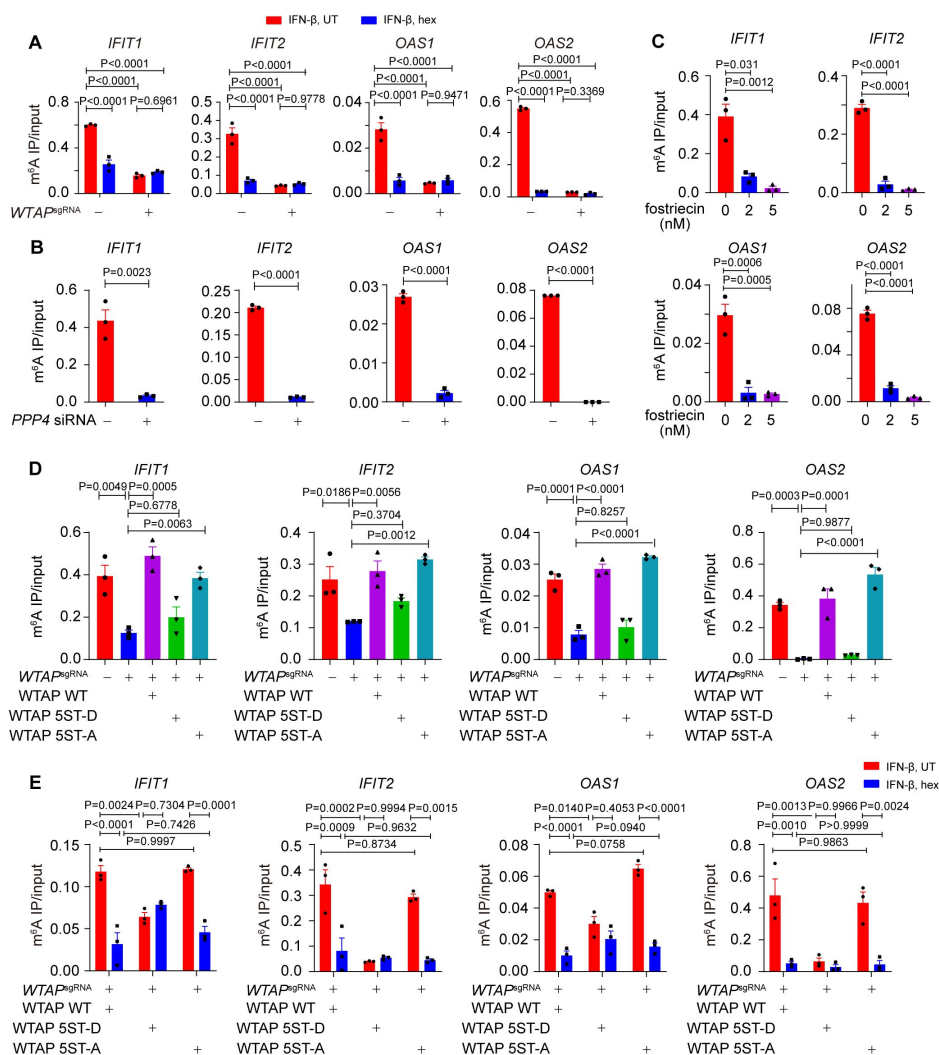


Figure 4.

Liquid-phase separation of WTAP mediates m⁶A modification and ISGs expression.

(A) Control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages were treated with 10 ng/mL IFN-β together with or without 5% 1,6-hexanediol (hex) and 20 μg/mL digitonin for 4 hours or left untreated. MeRIP-qPCR analysis of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A-modified mRNA and input were shown (m⁶A IP/input).

(B) THP-1-derived macrophages transfected with *scramble* (*scr*) siRNA or *PPP4* siRNA were treated with 10 ng/mL IFN-β for 4 hours. MeRIP-qPCR assay of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A-modified mRNA and input were shown (m⁶A IP/input).

(C) THP-1-derived macrophages were pre-treated with 2 nM or 5 nM fostriecin for 24 hours or left untreated, followed with 10 ng/mL IFN-β for 4 hours. MeRIP-qPCR analysis of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A-modified mRNA and input were shown (m⁶A IP/input).

(D) Control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages with or without expression of Flag-tagged wild type (WT) WTAP, and its 5ST-D or 5ST-A mutants were treated with 10 ng/mL IFN-β for 4 hours. MeRIP-qPCR assay of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A-modified mRNA and input were shown (m⁶A IP/input).

(E) Wild type (WT) WTAP, 5ST-D or 5ST-A mutant-rescued *WTAP^{sgRNA}* THP-1-derived macrophages were treated with 10 ng/mL IFN-β for 4 hours. MeRIP-qPCR assay of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A-modified mRNA and input were shown (m⁶A IP/input).

All error bars, mean values ± SEM, P-values were determined by two-way ANOVA test of *n* = 3 independent biological experiments in (A, E). All error bars, mean values ± SEM, P-values were determined by unpaired two-tailed Student's *t*-test of *n* = 3 independent biological experiments in (B-C). All error bars, mean values ± SEM, P-values were determined by one-way ANOVA test of *n* = 3 independent biological experiments in (D).

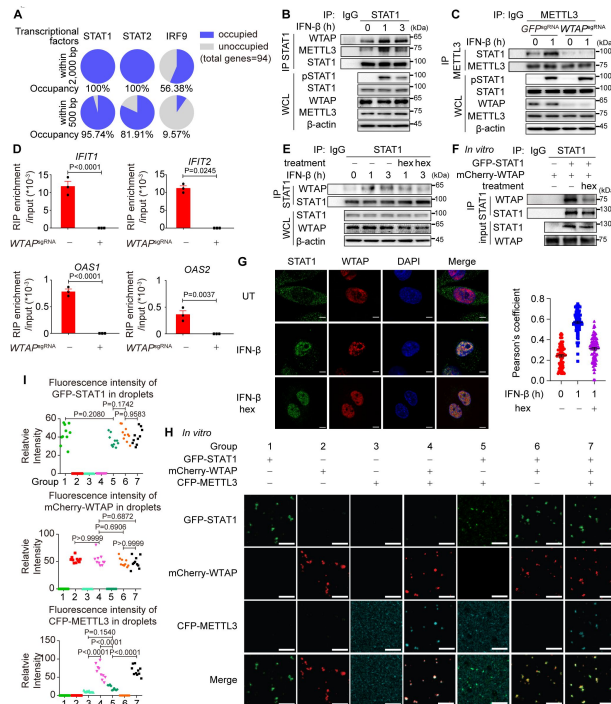


Figure 5.

Liquid-phase separated WTAP bridges STAT1 and m⁶A modification methyltransferase complex to direct m⁶A modification on ISG mRNAs.

(A) Occupancy of the promoter of the WTAP down-regulated ISGs with lower m⁶A level in *WTAP*^{sgRNA} #2 cells (genes were identified in **Figure 3A** and both -2000-0 bp and -500-0 bp upstream of transcription start site were analyzed) by STAT1, STAT2 and IRF9 was predicted by AnimalTFDB 3.0.

(B) THP-1-derived macrophages were treated with 10 ng/mL IFN- β for indicated time points. Whole cell lysate (WCL) was collected and immunoprecipitation (IP) experiment using anti-STAT1 antibody or rabbit IgG was performed, followed with immunoblot.

(C) Control (*GFP*^{sgRNA}) and *WTAP*^{sgRNA} #2 THP-1-derived macrophages were treated with 10 ng/mL IFN- β for 1 hour or left untreated. WCL was collected and IP experiment using anti-METTL3 antibody or rabbit IgG was performed, followed with immunoblot.

(D) Control (*GFP*^{sgRNA}) and *WTAP*^{sgRNA} #2 THP-1-derived macrophages were treated with 10 ng/mL IFN- β for 2 hour or left untreated. RNA immunoprecipitation (RIP) experiments using anti-METTL3 antibody or rabbit IgG control were performed in control and *WTAP*^{sgRNA} #2 THP-1-derived macrophages treated with 10 ng/mL IFN- β for 2 hours. RNA of ISGs enriched by RIP was analyzed by quantitative real time-polymerase chain reaction (qPCR) assay, and the ratios between RIP-enriched RNA and input were shown (RIP enrichment/input).

(E) THP-1-derived macrophages were treated with 10 ng/mL IFN- β for indicated time points together with or without 5% 1,6-hexanediol (hex) and 20 μ g/mL digitonin. WCL was collected and IP experiment with anti-STAT1 antibody or rabbit IgG was performed, followed with immunoblot.

(F) Recombinant GFP-STAT1 (10 μ M) and mCherry-WTAP (10 μ M) were incubated with or without 5% hex *in vitro*. IP experiment with anti-STAT1 antibody or rabbit IgG was performed, followed with immunoblot.

(G) THP-1-derived macrophages were treated with 10 ng/mL IFN- β only or with 5% hex and 20 μ g/mL digitonin for 1 hour or left untreated (UT). Interaction between STAT1 (green) and WTAP (red) were imaged using confocal microscope. Pearson's correlation coefficient was analyzed and calculated from $n = 80$ cells through ImageJ. Scale bars indicated 5 μ m.

(H-I) Recombinant GFP-STAT1 (10 μ M), mCherry-WTAP (10 μ M) and CFP-METTL3 (10 μ M) were incubated using physiological buffer at 37°C *in vitro*. After incubation, images were captured using confocal microscope (H). Relative fluorescence intensity of proteins in $n = 10$ condensates were analyzed by ImageJ software (I).

All error bars, mean values $\pm$ SEM, P-values were determined by unpaired two-tailed Student's *t*-test of $n = 3$ independent biological experiments in (D). All error bars, mean values $\pm$ SD, P-values were determined by unpaired two-tailed Student's *t*-test of $n = 80$ cells in (G) and $n = 3$ independent biological experiments in (I). For (B-D, E-G and H), similar results were obtained for three independent biological experiments.

but not its de-phosphorylation event (Figure 5-figure supplement 1H). Altogether, our data showed that phase separated WTAP functioned as the key adaptor protein bridging METTL3 and STAT1 to direct the m⁶A modification of ISG mRNAs.

We next tried to figure out how STAT1-MTC bound and mediated m⁶A modification on ISG mRNAs. By performing the chromosome immunoprecipitation (ChIP)-qPCR experiments, we found that WTAP interacted with promoter regions of ISGs along with STAT1 under IFN- β stimulation (**Figure 6A-B**). STAT1 deficiency abolished the binding affinity between WTAP and ISG promoter regions (**Figure 6C**), implying that nucleus-translocated transcription factor STAT1 directed WTAP and MTC to promoter regions of ISGs to conduct m⁶A deposition process on mRNA. *PPP4* knockdown impaired interaction between WTAP and ISG promoter region, linking the interaction efficiency to phase transition of WTAP (**Figure 6C**). WTAP 5ST-D mutant THP-1 cells exhibited significantly reduced binding ability with ISG promoter regions compared to WT WTAP and 5ST-A mutant THP-1 cells after IFN- β stimulation (**Figure 6D**), which consistent with the m⁶A modification results (**Figure 4D**). Hex treatment also restrained the interaction between WTAP and ISGs promoter regions in WT WTAP and WTAP 5ST-A cells (**Figure 6E**), consistent with the m⁶A modification results (**Figure 4E**). Collectively, WTAP phase separation was important for the promoter regions targeting and subsequent m⁶A deposition on ISG mRNAs.

Taken together, our findings revealed the driving force of aggregation and liquid condensates formation of WTAP, and demonstrated the distinct function between aggregates and liquid-phase separated WTAP. Based on our findings, we proposed a working model of WTAP phase transition in the regulation of m⁶A modification of ISG mRNAs under IFN- β stimulation. In untreated cells, hyperphosphorylated WTAP tended to form gel-like aggregates with lower mobility, restricting interaction between WTAP-dependent MTC and ISG promoter region. After IFN- β stimulation, protein phosphatase *PPP4* mediated the dephosphorylation of WTAP, resulting in the phase transition from aggregates to liquid-phase of WTAP. Liquid-phase WTAP exhibited enhanced mobility, recruiting METTL3 and nucleus-translocated STAT1 to form STAT1-WTAP-METTL3 condensates. Guided by STAT1, these condensates were able to bind with ISG promoter regions and mediate the m⁶A modification of ISG mRNAs during the STAT1-driven transcription process, thereby prompting diverse regulation on stability of ISG mRNAs. Our work uncovered how WTAP phase transition couples gene transcription with mRNA modification, providing a novel mechanism on multi-layer regulation of ISGs expression via m⁶A modification (**Figure 6F**).

Discussion

Type I IFN response is a crucial part of the antiviral immunity, which is strictly controlled by various types of regulation to avoid unexpected tissue damages. m⁶A deposition is one of the most widespread post-transcriptional modifications, which is involved in various aspects of RNA metabolism to regulate different biological events, including antiviral response (Garcias Morales & Reyes, 2021). m⁶A deposition on IFN- β mRNA is reported to dampen type-I IFN production by destabilizing transcripts, yet no m⁶A is detected on ISG mRNAs in the same study (Winkler et al., 2019). Subsequent works, however, reveal the diverse effects on antiviral responses mediated by the m⁶A modification of ISG mRNAs. hnRNP2B1-dependent m⁶A modification of *cGAS*, *IFI16* and *STING* promotes their nuclear exportation to amplify the cytoplasmic innate sensor signaling, while METTL3/METTL14-mediated m⁶A modification on IFITM1 and MX1 up-regulates their translation efficiency to positively regulate the antiviral response (McFadden et al., 2021; L. Wang, Wen, & Cao, 2019). Conversely, m⁶A modification destabilizes *IRF7* mRNA to suppress antiviral responses (A. Wang et al., 2022). Virus hijacks MTC components (e.g., WTAP degradation) to suppress the translational of *IRF3* and stability of *IFNAR1* mRNA via reduced m⁶A deposition, thereby exacerbating antiviral dysregulation (Ge et al., 2021). In our study, we employed IFN- β treatment to avoid the influence of m⁶A modification-mediated IFN- β mRNA decay, experimentally validated m⁶A deposition on a subset of ISG mRNAs

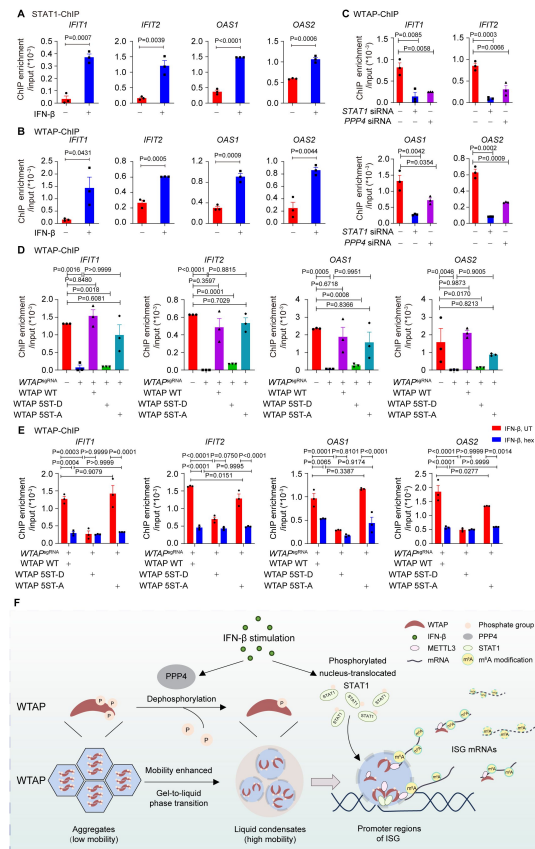


Figure 6.

Phase transition of WTAP mediates its interaction with ISGs promoter regions to regulate m⁶A modification on ISG mRNAs.

(A) Chromatin immunoprecipitation (ChIP) experiments using anti-STAT1 antibody or rabbit IgG control were performed in THP-1-derived macrophages treated with 10 ng/mL IFN- β for 2 hours or left untreated. Binding between the promoter regions of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* with WTAP was detected by quantitative real time-polymerase chain reaction (qPCR) assay. Ratios between ChIP-enriched DNA and input were shown (ChIP enrichment/input).

(B) ChIP experiments using anti-WTAP antibody or rabbit IgG control were performed in THP-1-derived macrophages treated with 10 ng/mL IFN- β for 2 hours or left untreated. Binding between the promoters of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* with WTAP were detected by qPCR assay. Ratios between ChIP-enriched DNA and input were shown (ChIP enrichment/input).

(C) ChIP experiments using anti-WTAP antibody or rabbit IgG control were performed in THP-1-derived macrophages transfected with *scramble* (*scr*) siRNA, *STAT1* siRNA or *PPP4* siRNA treated with 10 ng/mL IFN- β for 2 hours. Binding between the promoter regions of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* with WTAP was detected by qPCR assay. Ratios between ChIP-enriched DNA and input were shown (ChIP enrichment/input).

(D) ChIP experiments using anti-WTAP antibody or rabbit IgG control were performed in control (*GFP*^{sgRNA}) and *WTAP*^{sgRNA} #2 THP-1-derived macrophages with or without expression of Flag-tagged wild type (WT) WTAP, and its 5ST-D or 5ST-A mutants treated with 10 ng/mL IFN- β for 2 hours. Binding between the promoter regions of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* with WTAP was detected by qPCR assay. Ratios between ChIP-enriched DNA and input were shown (ChIP enrichment/input).

(E) ChIP experiments using anti-WTAP antibody or rabbit IgG control were performed in WT WTAP, 5ST-D or 5ST-A mutant-rescued *WTAP*^{sgRNA} THP-1-derived macrophages treated with 10 ng/mL IFN- β along with 5% 1,6-hexanediol (hex) for 2 hours. Binding between the promoters of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* with WTAP were detected by qPCR assay. Ratios between ChIP-enriched DNA and input were shown (ChIP enrichment/input).

(F) Schematic figure of IFN- β -induced phase transition of WTAP regulates the m⁶A modification of ISG mRNAs. All error bars, mean values $\pm$ SEM, P-values were determined by unpaired two-tailed Student's *t*-test of *n* = 3 independent biological experiments in (A-C). All error bars, mean values $\pm$ SEM, P-values were determined by one-way ANOVA test of *n* = 3 independent biological experiments in (D). All error bars, mean values $\pm$ SEM, P-values were determined by two-way ANOVA test of *n* = 3 independent biological experiments in (E).

fine-tunes antiviral responses by reducing transcript stability, maintaining a steady antiviral response and alleviates the hyper-activation of IFN- β signaling. Our findings together demonstrated the diverse function of m⁶A modification on ISG mRNAs, including suppression of transcription, promotion of translation, regulation of mRNA stabilization or even the cellular trafficking of mRNA, highlighting the complexity of m⁶A modification in mRNA fate determination, and emphasizing the needs of precise regulation on immune response.

WTAP, a core scaffolding protein of MTC, recruits and stabilizes other MTC components, such as METTL3 and METTL14, into nuclear speckles. Abnormal expression of WTAP is discovered in multiple diseases, including cancer or diabetes, to address dysfunctional m⁶A modification in specific genes (Chen et al., 2019 [DOI](#); Q. Huang, Mo, Liao, Chen, & Zhang, 2022 [DOI](#); Z. X. Li et al., 2022 [DOI](#); Yu et al., 2021 [DOI](#)). However, how WTAP reacts in response to specific stimulation and selectively targets specific mRNA remains unclear. Here, we focused on the function of WTAP on antiviral response, uncovered the novel mechanism that WTAP underwent IFN- β -induced dephosphorylation-mediated gel-to-liquid phase transition, which mediated its interaction with transcription factor STAT1, directing MTC to conduct the m⁶A modification on nascent ISG mRNAs co-transcriptionally.

Phase separation is a well-known phenomenon associated with the formation of membraneless organelles, which concentrates the substrates and enzymes and accelerates biochemistry reactions (Alberti, Gladfelter, & Mittag, 2019 [DOI](#)). Previous researches uncover the involvement of phase separation in m⁶A modification process, METTL3 and YTHDFs undergo LLPS during the m⁶A deposition and recognition process (Han et al., 2022 [DOI](#); J. Li et al., 2022 [DOI](#)), and m⁶A modification promotes the LLPS of RNA-binding proteins, including m⁶A-mRNA-YTHDFs and m⁶A-eRNA/YTHDC1/BRD4 condensates (Gao et al., 2019 [DOI](#); Lee et al., 2021 [DOI](#); Ries et al., 2019 [DOI](#); J. Wang et al., 2020 [DOI](#)). However, whether other MTC components exhibits phase separation is still unknown. Here, we observed that WTAP also went through phase separation both *in vitro* and *in vivo*. We found WTAP mainly formed the aggregates with lower mobility in untreated cells, and transited to liquid condensates with IFN- β treatment, implying that phase separation participated in WTAP-dependent m⁶A deposition under IFN- β stimulation. We found hex treatment or phospho-mimic mutations (5ST-D) trapped WTAP in low-mobility aggregates, impairing ISG promoter binding and m⁶A deposition. These results uncovered the mechanism and the crucial role of WTAP phase separation and phase transition in MTC assembly and ISGs m⁶A deposition under IFN- β stimulation. Furthermore, the phase transition of WTAP suggested a potential mechanism that aggregation of WTAP might work as a backup storage, and the stimuli-induced phase transition was developed to perform the m⁶A modification timely and precisely in response to the requirement of cell development or abnormal situations. Thus, investigation of WTAP phase separation in different model could be conducted to explain the mechanism of m⁶A modification under complex stimulation and explore possible therapeutic targets of multiple diseases.

Post-translational modifications are one of the most common ways to modulate characteristic of proteins (Y. Y. Luo, Wu, & Li, 2021 [DOI](#)), like phosphorylation of serine and threonine (Markevich, Hoek, & Kholodenko, 2004 [DOI](#)) and methylation of arginine (Guccione & Richard, 2019 [DOI](#)). Several reports indicate that phosphorylation of phase-separated proteins are associated with the solid/gel-like aggregates formation or solid/gel-to-liquid phase transition, participating in various physiological process such as the pathogenesis of tau protein aggregates or acute transcriptional response (Arendt, Stieler, & Holzer, 2016 [DOI](#); Boyko et al., 2019 [DOI](#); H. Zhang et al., 2022 [DOI](#)), but whether the phosphorylation-controlled phase transition of proteins regulate other physiological process remains to be illustrated. In this study, we observed PPP4-mediated dephosphorylation of WTAP at serine/threonine residues triggered a gel-to-liquid phase transition, enabling STAT1-MTC recruitment. Consistently, phosphomimic 5ST-D mutant of WTAP mainly aggregated as lower mobility form compared to phosphodeficient 5ST-A mutant of WTAP, which formed liquid condensates in cells, indicating that phosphorylation level of WTAP balanced its phase transition. Therefore, our findings shed light on the mechanism of post-translational modification-regulated

phase transition of WTAP and put forward an open question whether phosphorylation and other post-translational modifications function as the key strategy for global phase transition of proteins *in vivo*.

Since we found the phase transition event of WTAP and its interaction with transcription factor STAT1, we proposed that co-transcriptional m⁶A modification on nascent ISG transcripts might occur during IFN- β stimulation. As a global modification in cells, m⁶A modification can be hindered by MTC along with the inhibition role of exon junction complex, lead to the specificity of modification, mapping the m⁶A topology and maintain in stable state (He et al., 2023 [DOI](#); Z. Luo et al., 2023 [DOI](#); Uzonyi et al., 2023 [DOI](#); Yang, Triboulet, Liu, Sendinc, & Gregory, 2022 [DOI](#)). More and more results suggest that MTC can be recruited by methylated histone proteins or transcription factors, enrich in different chromatin loci, such as promoter, to dynamically deposit m⁶A modification on nascent mRNA co-transcriptionally under certain circumstance or function in various physiological activities, implying the importance of interaction between MTC and chromatin architecture (Dou et al., 2023 [DOI](#); Sendinc & Shi, 2023 [DOI](#); Xu et al., 2022 [DOI](#)). Transcription factor SMAD2/3, STAT5B, KLF9 and CEBPZ are found to recruit MTC on transcriptional start sites (TSS) and specifically direct the m⁶A modification on a subset of downstream transcripts mRNA CDS region, thereby affecting the cell fate decision (Barbieri et al., 2017 [DOI](#); Bertero et al., 2018 [DOI](#); Bhattarai et al., 2024 [DOI](#)). In line with this, we validated that the IFN- β -induced WTAP-STAT1 interaction guided MTC to the promoter regions of ISGs to direct the m⁶A modification on ISG mRNAs. We also revealed that the WTAP-mediated m⁶A modification of ISG mRNAs mainly occurs within the coding regions, different from its distribution in the general transcriptome that enriched within mRNA 3'-UTR. Our findings raised a novel perspective that transcription factor-MTC interaction directed co-transcriptional m⁶A modification to balance immune activation and homeostasis.

Taken together, our study drew a full picture of the phase transition of WTAP and the function of STAT1-MTC directed m⁶A modification on ISG mRNAs under IFN- β stimulation. Our findings unraveled a novel mechanism of fine-tuning m⁶A methylated mRNA profile under stimulation, and provided a possible therapeutic target in antiviral responses and many other diseases.

Materials and Methods

Cells

THP-1 cells obtained from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China) were cultured in RPMI 1640 (Gibco) with 10% (vol/vol) fetal bovine serum (FBS), 1% penicillin-streptomycin (Gibco, 1:100) and 1% L-glutamine (Gibco) incubated in a 37°C chamber with 5% CO₂ (Thermo Fisher Scientific). Before stimulation, THP-1 cells were differentiated into macrophages (THP-1-derived macrophages) through treatment of 100 nM phorbol-12-myristate-13-acetate (PMA) (P8139, Sigma) for 16 hours. After PMA treatment, the macrophages were rested for 48 hours before stimulation. A549 and HeLa cells obtained from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China) were cultured in DMEM medium (Corning) with 10% FBS and 1% L-glutamine (Gibco) incubated in a 37°C chamber with 5% CO₂.

Reagents and Antibodies

IFN- β (cat. #300-02BC) was purchased from PeproTech Inc. PMA (cat. #P1585), puromycin (cat. #P9620), Actinomycin D (cat. #SBR00013), 1,6-hexanediol (cat. #240117), digitonin (cat. #D141), DTT (cat. #3483-12-3), DAPI (cat. #D9542) were purchased from Sigma-Aldrich. Fostriecin (cat. #ab144255) was purchased from Abcam. Isopropyl-beta-D-thiogalactopyranoside (IPTG, cat. #CA413) and Superluciferase High-efficiency Transfection Reagent (cat. # 11231804) were purchased

from MIKX. Hieff-Trans Liposomal Transfection Reagent (40802ES02) were purchased from Yeasen. NP-40 (cat. #P0013F) was purchased from Beyotime Biotechnology. The antibodies that used in this study were listed in *Supplementary Appendix, Table S1*.

Generation of knockout cell lines by CRISPR/Cas9

After THP-1 cells were seeded, medium was replaced by DMEM containing polybrene (8 µg/ml) (Sigma-Aldrich) lentiviral vector encoding Cas9 and small guide RNAs (sgRNA) for 48 hours. Cells were selected using puromycin (Sigma-Aldrich). The sequence of sgRNAs targeting indicated gene obtained from Sangon (Shanghai, China) were listed in *Supplementary Appendix, Table S2*:

siRNA Transfection

siRNA duplexes and scrambled siRNA were chemically synthesized by RIBOBIO (Guangzhou, China) and transfected into cells using LipoRNAiMAX reagent (Invitrogen) according to the manufacturer's instructions. The sequences of siRNAs were listed in *Supplementary Appendix, Table S2*:

Virus infection

VSV and HSV-1 were amplified and tittered on Vero E6 cell line. Virus titers were measured by means of 50% of the tissue culture's infectious dose (TCID₅₀). Virus were infected into indicated cell lines with indicated titers as shown in figure legends.

Immunoblot (IB) assays and immunoprecipitation (IP)

Relevant cells were stimulated as indicated, and whole cell lysate was obtained using low-salt lysis buffer (50 mM Hepes (pH 7.5), 150 mM NaCl, 1 mM EDTA, 1.5 mM MgCl₂, 10% glycerol, 1% Triton X-100, supplemented with protease inhibitor cocktail (cat. #A32965, Pierce) and phosSTOP Phosphatase Inhibitor Cocktail (cat. #4906837001, Roche)). Cell lysates were centrifuged at 4°C, 12000 g for 15 minutes and supernatants was boiled at 100 °C for 5 minutes with the 5×Loading Buffer (cat. #FD006, Hangzhou Fude Biological Technology Co, LTD.). Solution was resolved by SDS-PAGE, transferred to polyvinylidene fluoride (PVDF) membranes (cat. #1620177, Bio-Rad Laboratories, Inc.), blocked with 5% skim milk (cat. #232100, BD) for 1 hour and incubated with appropriate antibody. Immobilon Western HRP Substrate (cat. #WBKLS0500, Millipore) was used for protein detection with ChemiDoc MP System (Bio-Rad Laboratories, Inc.) and Image Lab version 6.0 (Bio-Rad software, California, USA). For immunoprecipitation, protein samples were incubated with protein A/G beads (cat. #20333, #20399, Pierce) or anti-Flag M2 affinity gel (cat. #A2220, Sigma-Aldrich) together with indicated antibody at 4°C overnight. Beads were washed five times with low-salt lysis buffer and boiled at 100 °C for 5 minutes with 2×SDS loading buffer, followed with immunoblot assays described above.

Mass spectrometry

HEK 293T cells were transfected with plasmids expressing mCherry-WTAP. 24 hours after transfection, whole cell lysate was obtained using low-salt lysis buffer and immunoprecipitated using protein A/G beads together with anti-WTAP antibody at 4°C overnight. Proteins were obtained from beads and analyzed by LC-MS/MS analysis. Details were shown in *Supplementary Appendix, Extended Methods*.

RNA extraction, quantitative real-time polymerase chain reaction (qPCR) and RNA sequencing (RNA-Seq) assay

Relevant cells were treated as indicated and total RNA was extracted from cells with TRIzol reagent (cat. #15596026, Invitrogen), according to the manufacturer's instructions. RNA was then reverse-transcribed into cDNA using HiScript[®] III RT SuperMix for qPCR (+gDNA wiper) (cat.

#R323-01, Vazyme). The level of indicated genes were determined by qPCR with 2×PolarSignal™ SYBR Green mix Taq (cat. #MKG900-10, MIKX) using LightCycler® 480 System (Roche). The primers used in qPCR were listed in *Supplementary Appendix, Table S2*. For RNA-Seq assay, total RNA was isolated from cells using TRIzol reagent, and sequencing was performed by Sangon Biotech (Shanghai, China). Details of RNA-Seq data analysis were shown in *Supplementary Appendix, Extended Methods*.

Chromatin-immunoprecipitation (ChIP) assay

THP-1 cells were treated as indicated and DNA were enriched through ChIP assay referred to the Rockland Chromatin Immunoprecipitation assay protocol. Details were shown in *Supplementary Appendix, Extended Methods*. After ChIP enrichment, DNA were analyzed through qPCR assay. The sequences of primers were listed in *Supplementary Appendix, Table S2*:

RNA-immunoprecipitation (RIP) followed with qPCR assay

THP-1 cells were treated as indicated and RNA were enriched through RIP assay referred to the previously reported protocol (Keene, Komisarow, & Friedersdorf, 2006 [\[3\]](#)). Details were shown in *Supplementary Appendix, Extended Methods*. After RIP enrichment, RNA was analyzed through qPCR assay. The sequences of primers were listed in *Supplementary Appendix, Table S2*:

m⁶A methylated RNA immunoprecipitation (MeRIP) followed with qPCR assay and MeRIP-Seq assay

For MeRIP-qPCR, the procedures were referred to the protocol as manufacturer's instructions. Approximately 2×10^7 cells were seeded and stimulated by 10 ng/mL IFN- β for 6 hours. Approximately more than 50 μ g of total RNA was subjected to isolate poly (A) mRNA with poly-T oligo attached magnetic beads (cat. #61006, Invitrogen). Following purification, the poly(A) mRNA fractions is fragmented into ~100-nt-long oligonucleotides using divalent cations under elevated temperature. Then the cleaved RNA fragments were subjected to incubation for 2h at 4°C with m⁶A-specific antibody (cat. #202003, Synaptic Systems) in IP buffer (50[mM Tris-HCl, 750[mM NaCl and 0.5% Igepal CA-630) supplemented with BSA (0.5[μ g/ μ L). The mixture was then incubated with protein-A beads and eluted with elution buffer (1×IP buffer and 6.7mM m⁶A), while 10% of fragmented mRNA was kept as input. After immunoprecipitation, precipitates were washed by IP buffer for 5 times and RNA was isolated by adding TRIzol reagent. Purified m⁶A-containing mRNA fragments and untreated mRNA input control fragments are converted to final cDNA library in accordance with a strand-specific library preparation by dUTP method, and analyzed by qPCR assay. The primers used in qPCR were listed in *Supplementary Appendix, Table S2*.

For MeRIP-Seq, approximately 5×10^7 cells were seeded and stimulated by 10 ng/mL IFN- β for 4 hours. After stimulation, total RNA was extracted using TRIzol reagent. The total RNA quality and quantity were analysis of Bioanalyzer 2100 and RNA 6000 Nano LabChip Kit (Agilent, CA, USA) with RIN number >7.0, followed with MeRIP assay. And then we performed the paired-end 2×150 bp sequencing on an Illumina Novaseq 6000 platform at the LC-BIO Bio-tech ltd (Hangzhou, China) following the vendor's recommended protocol. Details of MeRIP-Seq data analysis were shown in *Supplementary Appendix, Extended Methods*.

Immunofluorescence staining (IF) and imaging

THP-1-derived macrophages were differentiated and stimulated as described above. After stimulation, cells were fixed by in 4% paraformaldehyde (Meilunbio, China) for 15 minutes at room temperature, permeated by ice-cold methanol at -20 °C for 15 minutes and blocked using 6% goat serum (cat. #AR1009, Boster Biological) for 1 hour, and then incubated with indicated primary antibody at 4°C overnight, followed by incubation with indicated secondary antibody at room temperature for 1 hour. Phosphate buffer (PBS) was used to wash three times for 5 minutes

between each step. After staining, Leica TCS-SP8 STED 3X confocal fluorescence microscope was used to acquire images. Images were captured by Leica TCS-SP8 STED 3X confocal fluorescence microscope and analyzed by Leica Application Suite Advanced Fluorescence (LAS AF, Version 4.2) software, while deconvolution was conducted by Huygens Software (Version 23.04). The colocalization between proteins was quantified by ImageJ software (National Institutes of Health, Maryland, USA, Version 1.52) using Pearson's correlation coefficient, ranging from -1 to 1. A value of -1 represents perfect negative correlation, 0 means no correlation and 1 indicates perfect correlation.

Expression, purification, and interaction of mCherry-WT WTAP, 5ST-A/ST-D mutants, CFP-METTL3 and GFP-STAT1

Procedures of the expression of proteins in *E. coli* were referred to Hao Jiang et al. (Jiang et al., 2015). Recombinant mCherry-WTAP, CFP-METTL3 and GFP-STAT1 were cloned into pET-28A plasmid, while ST-D and 5ST-A mutants of WTAP were constructed by mutating the pET-28A-mCherry-WTAP plasmid using Muta-direct™ Kit (Sbsgene, cat. #SDM-15). After transformed into *E. coli*, protein was extracted as shown in *Supplementary Appendix, Extended Methods*.

For detection of interaction among mCherry-WTAP or its indicated mutants, CFP-METTL3 and GFP-STAT1, purified proteins were mixed using physiological buffer (20 mM Tris-HCl, pH 7.5, 150 mM KCl, 10% PEG8000) in the final concentration of 10 μ M and placed on the slides. Mixtures were incubated and imaged at 37°C in a live-cell-imaging chamber of Leica TCS-SP8 STED 3X confocal fluorescence microscope while images were analyzed by LAS AF software.

Phase separation assay and fluorescence recovery after photobleaching (FRAP)

Expression, purification, and interaction of mCherry-WT WTAP, 5ST-A/ST-D mutants, CFP-METTL3 and GFP-STAT1 were shown in *Supplementary Appendix, Extended Methods*. For *in vitro* phase separation assay, recombinant mCherry-WT WTAP, NTD, CTD, Δ PLD, 5ST-D and 5ST-A mutants in indicated concentration were mixed with physiological buffer (20 mM Tris-HCl, pH 7.5, 150 mM KCl, 10% PEG8000). After incubation, turbid solution in the 200 μ L PCR tubes were directed observed. For further capture of fluorescence images, solution of recombinant mCherry-WTAP and physiological buffer were treated with or without hex at a final concentration of 10% and placed on dishes. Leica TCS-SP8 STED 3X confocal fluorescence microscope with live-cell-imaging chamber at 37°C was used to capture fluorescence images and the 3D images. For *in cells* separation assay, plasmids expression mCherry-WT WTAP, 5ST-D and 5ST-A were transfected into HeLa cells for 24 hours, and treated with or without 5% hex and 20 μ g/mL digitonin for 1 hour. Leica TCS-SP8 STED 3X confocal fluorescence microscope with live-cell-imaging chamber at 37°C was used to capture fluorescence images. For observing the fusion of condensates, recombinant mCherry-WTAP mixed with physiological buffer or HeLa cells transfected with mCherry-WTAP plasmids were incubated in the live-cell-imaging chamber at 37°C and. Images were captured by Leica TCS-SP8 STED 3X confocal fluorescence microscope, while deconvolution was conducted by Huygens software.

For *in vitro* FRAP experiments were performed on Leica TCS-SP8 STED 3X confocal fluorescence microscope. Recombinant mCherry-WT WTAP, NTD, CTD, 5ST-D and 5ST-A mutants in indicated concentration was mixed with physiological buffer on dishes. For *in cells* FRAP experiment, HeLa cells transfected with mCherry-WT WTAP, 5ST-D and 5ST-A mutants were treated as indicated. Samples were imaged with Leica TCS-SP8 STED 3X confocal fluorescence microscope with live-cell-imaging chamber at 37°C. Spots of mCherry-WTAP foci were photobleached with 100% laser power using 567-nm lasers followed with time-lapse images. Images were analyzed by LAS AF software. Fluorescence intensities of regions of interest (ROIs) were corrected by unbleached control regions and then normalized to pre-bleached intensities of the ROIs.

Statistical Analysis

The data of all quantitative experiments are represented as mean $\pm$ SEM of three independent experiments. Statistical analyses were performed with GraphPad Prism software version 8.0 (GraphPad software, California, USA) using one/two-way analysis of variance (ANOVA) followed by Tukey's test or unpaired two-tailed Student's *t*-test as described in the figure legends. P-value < 0.05 was considered as statistically significant of all statistical analyses.

Data availability

Raw data of RNA-Seq and MeRIP-Seq have been deposited in the Sequence Read Archive (SRA) database under accession code PRJNA680033 and PRJNA680974, respectively. The data that support this study are available within the article and its Supplementary Information files or available from the authors upon request.

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Additional information

Author Contributions

J.C. conceived the project. *J.C.*, *S.C.* and *J.Z.* designed the experiments. *S.C.*, *J.Z.* and *S.J.* expressed and purified the proteins using *E. coli* system while *S.C.*, *J.Z.* and *C.Z.* finished the rest of the experiments. *X.L.*, *J.R.*, *S.C.* and *J.Z.* analyzed the data of RNA-Seq and MeRIP-Seq. All authors discussed the results and commented on or prepared the manuscript.

Additional files

Supplementary file [🔗](#)

References

- Alberti S., Gladfelter A., Mittag T (2019) **Considerations and Challenges in Studying Liquid-Liquid Phase Separation and Biomolecular Condensates** *Cell* **176**:419–434 <https://doi.org/10.1016/j.cell.2018.12.035>Google Scholar
- Arendt T., Stieler J. T., Holzer M (2016) **Tau and tauopathies** *Brain Res Bull* **126**:238–292 <https://doi.org/10.1016/j.brainresbull.2016.08.018>Google Scholar
- Barbieri I., Tzelepis K., Pandolfini L., Shi J., Millan-Zambrano G., Robson S. C., ..., Kouzarides T. (2017) **Promoter-bound METTL3 maintains myeloid leukaemia by m(6)A-dependent translation control** *Nature* **552**:126–131 <https://doi.org/10.1038/nature24678>Google Scholar
- Bertero A., Brown S., Madrigal P., Osnato A., Ortmann D., Yiangou L., ..., Vallier L. (2018) **The SMAD2/3 interactome reveals that TGF β controls m6A mRNA methylation in pluripotency** *Nature* **555**:256–259 <https://doi.org/10.1038/nature25784>Google Scholar
- Bhattarai P. Y., Kim G., Lim S. C., Choi H. S (2024) **METTL3-STAT5B interaction facilitates the co-transcriptional m(6)A modification of mRNA to promote breast tumorigenesis** *Cancer Lett* **603**:217215 <https://doi.org/10.1016/j.canlet.2024.217215>Google Scholar
- Boyko S., Qi X., Chen T. H., Surewicz K., Surewicz W. K (2019) **Liquid-liquid phase separation of tau protein: The crucial role of electrostatic interactions** *J Biol Chem* **294**:11054–11059 <https://doi.org/10.1074/jbc.AC119.009198>Google Scholar
- Chen Y., Peng C., Chen J., Chen D., Yang B., He B., ..., Zheng S. (2019) **WTAP facilitates progression of hepatocellular carcinoma via m6A-HuR-dependent epigenetic silencing of ETS1** *Mol Cancer* **18**:127 <https://doi.org/10.1186/s12943-019-1053-8>Google Scholar
- Cui J., Song Y., Li Y., Zhu Q., Tan P., Qin Y., ..., Wang R. F (2014) **USP3 inhibits type I interferon signaling by deubiquitinating RIG-I-like receptors** *Cell Res* **24**:400–416 <https://doi.org/10.1038/cr.2013.170>Google Scholar
- Dou X., Xiao Y., Shen C., Wang K., Wu T., Liu C., ..., He C. (2023) **RBFOX2 recognizes N(6)-methyladenosine to suppress transcription and block myeloid leukaemia differentiation** *Nat Cell Biol* **25**:1359–1368 <https://doi.org/10.1038/s41556-023-01213-w>Google Scholar
- Gabryelczyk B., Cai H., Shi X., Sun Y., Swinkels P. J. M., Salentinig S., ..., Miserez A. (2019) **Hydrogen bond guidance and aromatic stacking drive liquid-liquid phase separation of intrinsically disordered histidine-rich peptides** *Nat Commun* **10**:5465 <https://doi.org/10.1038/s41467-019-13469-8>Google Scholar
- Gao Y., Pei G., Li D., Li R., Shao Y., Zhang Q. C., Li P (2019) **Multivalent m6A motifs promote phase separation of YTHDF proteins** *Cell Res* **29**:767–769 <https://doi.org/10.1038/s41422-019-0210-3>Google Scholar

- Garcias Morales D., Reyes J. L. (2021) **A birds'-eye view of the activity and specificity of the mRNA m6A methyltransferase complex** *Wiley Interdiscip Rev RNA* **12**:e1618 <https://doi.org/10.1002/wrna.1618>Google Scholar
- Ge Y., Ling T., Wang Y., Jia X., Xie X., Chen R., ..., Xu A. (2021) **Degradation of WTAP blocks antiviral responses by reducing the m6A levels of IRF3 and IFNAR1 mRNA** *EMBO Rep* **22**:e52101 <https://doi.org/10.15252/embr.202052101>Google Scholar
- Geula S., Moshitch-Moshkovitz S., Dominissini D., Mansour A. A., Kol N., Salmon-Divon M., ..., Hanna J. H (2015) **Stem cells. m6A mRNA methylation facilitates resolution of naive pluripotency toward differentiation** *Science* **347**:1002–1006 <https://doi.org/10.1126/science.1261417>Google Scholar
- Goubau D., Deddouche S., Reis e Sousa C. (2013) **Cytosolic sensing of viruses** *Immunity* **38**:855–869 <https://doi.org/10.1016/j.immuni.2013.05.007>Google Scholar
- Gourfinkel-An I., Cancel G., Trottier Y., Devys D., Tora L., Lutz Y., ..., Hirsch E. C (1997) **Differential distribution of the normal and mutated forms of huntingtin in the human brain** *Ann Neurol* **42**:712–719 <https://doi.org/10.1002/ana.410420507>Google Scholar
- Guccione E., Richard S (2019) **The regulation, functions and clinical relevance of arginine methylation** *Nat Rev Mol Cell Biol* **20**:642–657 <https://doi.org/10.1038/s41580-019-0155-x>Google Scholar
- Han D., Longhini A. P., Zhang X., Hoang V., Wilson M. Z., Kosik K. S (2022) **Dynamic assembly of the mRNA m6A methyltransferase complex is regulated by METTL3 phase separation** *PLoS Biol* **20**:e3001535 <https://doi.org/10.1371/journal.pbio.3001535>Google Scholar
- He P. C., Wei J., Dou X., Harada B. T., Zhang Z., Ge R., ..., He C. (2023) **Exon architecture controls mRNA m(6)A suppression and gene expression** *Science* **379**:677–682 <https://doi.org/10.1126/science.abj9090>Google Scholar
- Henault J., Riggs J. M., Karnell J. L., Liarski V. M., Li J., Shirinian L., ..., Sanjuan M. A (2016) **Self-reactive IgE exacerbates interferon responses associated with autoimmunity** *Nat Immunol* **17**:196–203 <https://doi.org/10.1038/ni.3326>Google Scholar
- Hou S., Hu J., Yu Z., Li D., Liu C., Zhang Y (2024) **Machine learning predictor PSPire screens for phase-separating proteins lacking intrinsically disordered regions** *Nat Commun* **15**:2147 <https://doi.org/10.1038/s41467-024-46445-y>Google Scholar
- Hu H., Miao Y. R., Jia L. H., Yu Q. Y., Zhang Q., Guo A. Y (2019) **AnimalTFDB 3.0: a comprehensive resource for annotation and prediction of animal transcription factors** *Nucleic Acids Res* **47**:D33–D38 <https://doi.org/10.1093/nar/gky822>Google Scholar
- Huang H., Weng H., Zhou K., Wu T., Zhao B. S., Sun M., ..., Chen J. (2019) **Histone H3 trimethylation at lysine 36 guides m(6)A RNA modification co-transcriptionally** *Nature* **567**:414–419 <https://doi.org/10.1038/s41586-019-1016-7>Google Scholar
- Huang Q., Mo J., Liao Z., Chen X., Zhang B (2022) **The RNA m6A writer WTAP in diseases: structure, roles, and mechanisms** *Cell Death Dis* **13**:852 <https://doi.org/10.1038/s41419-022-05268-9>Google Scholar
- Jia G., Fu Y., Zhao X., Dai Q., Zheng G., Yang Y., ..., He C. (2011) **N6-Methyladenosine in nuclear RNA is a major substrate of the obesity-associated FTO** *Nature Chemical Biology* **7**:885–

887 <https://doi.org/10.1038/nchembio.687>Google Scholar

Jiang H., Wang S., Huang Y., He X., Cui H., Zhu X., Zheng Y (2015) **Phase transition of spindle-associated protein regulate spindle apparatus assembly** *Cell* **163**:108–122 <https://doi.org/10.1016/j.cell.2015.08.010>Google Scholar

Keene J. D., Komisarow J. M., Friedersdorf M. B (2006) **RIP-Chip: the isolation and identification of mRNAs, microRNAs and protein components of ribonucleoprotein complexes from cell extracts** *Nat Protoc* **1**:302–307 <https://doi.org/10.1038/nprot.2006.47>Google Scholar

Knuckles P., Bühler M (2018) **Adenosine methylation as a molecular imprint defining the fate of RNA** *FEBS Letters* **592**:2845–2859 <https://doi.org/10.1002/1873-3468.13107>Google Scholar

Lee J. H., Wang R., Xiong F., Krakowiak J., Liao Z., Nguyen P. T., ..., Li W. (2021) **Enhancer RNA m6A methylation facilitates transcriptional condensate formation and gene activation** *Mol Cell* **81**:3368–3385 <https://doi.org/10.1016/j.molcel.2021.07.024>Google Scholar

Li J., Chen K., Dong X., Xu Y., Sun Q., Wang H., ..., Luo M. (2022) **YTHDF1 promotes mRNA degradation via YTHDF1-AGO2 interaction and phase separation** *Cell Prolif* **55**:e13157 <https://doi.org/10.1111/cpr.13157>Google Scholar

Li Z. X., Zheng Z. Q., Yang P. Y., Lin L., Zhou G. Q., Lv J. W., ..., Sun Y. (2022) **WTAP-mediated m6A modification of lncRNA DIAPH1-AS1 enhances its stability to facilitate nasopharyngeal carcinoma growth and metastasis** *Cell Death Differ* **29**:1137–1151 <https://doi.org/10.1038/s41418-021-00905-w>Google Scholar

Luo Y. Y., Wu J. J., Li Y. M (2021) **Regulation of liquid-liquid phase separation with focus on post-translational modifications** *Chem Commun (Camb)* **57**:13275–13287 <https://doi.org/10.1039/d1cc05266g>Google Scholar

Luo Z., Ma Q., Sun S., Li N., Wang H., Ying Z., Ke S (2023) **Exon-intron boundary inhibits m(6)A deposition, enabling m(6)A distribution hallmark, longer mRNA half-life and flexible protein coding** *Nat Commun* **14**:4172 <https://doi.org/10.1038/s41467-023-39897-1>Google Scholar

Markevich N. I., Hoek J. B., Kholodenko B. N (2004) **Signaling switches and bistability arising from multisite phosphorylation in protein kinase cascades** *J Cell Biol* **164**:353–359 <https://doi.org/10.1083/jcb.200308060>Google Scholar

McFadden M. J., McIntyre A. B. R., Mourelatos H., Abell N. S., Gokhale N. S., Ipas H., ..., Horner S. M (2021) **Post-transcriptional regulation of antiviral gene expression by N6-methyladenosine** *Cell Rep* **34**:108798 <https://doi.org/10.1016/j.celrep.2021.108798>Google Scholar

McFadden M. J., Sacco M. T., Murphy K. A., Park M., Gokhale N. S., Somfleth K. Y., Horner S. M (2022) **FTO Suppresses STAT3 Activation and Modulates Proinflammatory Interferon-Stimulated Gene Expression** *J Mol Biol* **434**:167247 <https://doi.org/10.1016/j.jmb.2021.167247>Google Scholar

- Meyer K. D., Jaffrey S. R (2017) **Rethinking m6A Readers, Writers, and Erasers** *Annu Rev Cell Dev Biol* **33**:319–342 <https://doi.org/10.1146/annurev-cellbio-100616-060758>Google Scholar
- Mowen K. A., Tang J., Zhu W., Schurter B. T., Shuai K., Herschman H. R., David M (2001) **Arginine methylation of STAT1 modulates IFN α /beta-induced transcription** *Cell* **104**:731–741 [https://doi.org/10.1016/s0092-8674\(01\)00269-0](https://doi.org/10.1016/s0092-8674(01)00269-0)Google Scholar
- Murthy A. C., Dignon G. L., Kan Y., Zerze G. H., Parekh S. H., Mittal J., Fawzi N. L (2019) **Molecular interactions underlying liquid-liquid phase separation of the FUS low-complexity domain** *Nat Struct Mol Biol* **26**:637–648 <https://doi.org/10.1038/s41594-019-0250-x>Google Scholar
- Oerum S., Meynier V., Catala M., Tisne C (2021) **A comprehensive review of m6A/m6Am RNA methyltransferase structures** *Nucleic Acids Res* **49**:7239–7255 <https://doi.org/10.1093/nar/gkab378>Google Scholar
- Perutz M. F (1996) **Glutamine repeats and inherited neurodegenerative diseases: molecular aspects** *Curr Opin Struct Biol* **6**:848–858 [https://doi.org/10.1016/s0959-440x\(96\)80016-9](https://doi.org/10.1016/s0959-440x(96)80016-9)Google Scholar
- Qin Y., Liu Q., Tian S., Xie W., Cui J., Wang R. F (2016) **TRIM9 short isoform preferentially promotes DNA and RNA virus-induced production of type I interferon by recruiting GSK3 β to TBK1** *Cell Res* **26**:613–628 <https://doi.org/10.1038/cr.2016.27>Google Scholar
- Reichheld S. E., Muiznieks L. D., Keeley F. W., Sharpe S (2017) **Direct observation of structure and dynamics during phase separation of an elastomeric protein** *Proc Natl Acad Sci U S A* **114**:E4408–E4415 <https://doi.org/10.1073/pnas.1701877114>Google Scholar
- Ries R. J., Zaccara S., Klein P., Olarerin-George A., Namkoong S., Pickering B. F., ..., Jaffrey S. R (2019) **m6A enhances the phase separation potential of mRNA** *Nature* **571**:424–428 <https://doi.org/10.1038/s41586-019-1374-1>Google Scholar
- Rubio R. M., Depledge D. P., Bianco C., Thompson L., Mohr I (2018) **RNA m6A modification enzymes shape innate responses to DNA by regulating interferon beta** *Genes Dev* **32**:1472–1484 <https://doi.org/10.1101/gad.319475.118>Google Scholar
- Schapira M (2016) **Structural Chemistry of Human RNA Methyltransferases** *ACS Chem Biol* **11**:575–582 <https://doi.org/10.1021/acscchembio.5b00781>Google Scholar
- Schneider W. M., Chevillotte M. D., Rice C. M (2014) **Interferon-stimulated genes: a complex web of host defenses** *Annu Rev Immunol* **32**:513–545 <https://doi.org/10.1146/annurev-immunol-032713-120231>Google Scholar
- Sendinc E., Shi Y (2023) **RNA m6A methylation across the transcriptome** *Mol Cell* **83**:428–441 <https://doi.org/10.1016/j.molcel.2023.01.006>Google Scholar
- Shi H., Wang X., Lu Z., Zhao B. S., Ma H., Hsu P. J., ..., He C. (2017) **YTHDF3 facilitates translation and decay of N(6)-methyladenosine-modified RNA** *Cell Res* **27**:315–328 <https://doi.org/10.1038/cr.2017.15>Google Scholar
- Tanaka M., Morishima I., Akagi T., Hashikawa T., Nukina N (2001) **Intra- and intermolecular beta-pleated sheet formation in glutamine-repeat inserted myoglobin as a model for polyglutamine diseases** *J Biol Chem* **276**:45470–45475 <https://doi.org/10.1074/jbc.M107502200>Google Scholar

- Uddin S., Sassano A., Deb D. K., Verma A., Majchrzak B., Rahman A., ..., Platanias L. C (2002) **Protein kinase C-delta (PKC-delta) is activated by type I interferons and mediates phosphorylation of Stat1 on serine 727** *Journal Of Biological Chemistry* **277**:14408–14416 <https://doi.org/10.1074/jbc.M109671200>Google Scholar
- Uzonyi A., Dierks D., Nir R., Kwon O. S., Toth U., Barbosa I., Schwartz S (2023) **Exclusion of m6A from splice-site proximal regions by the exon junction complex dictates m6A topologies and mRNA stability** *Mol Cell* **83**:237–251 <https://doi.org/10.1016/j.molcel.2022.12.026>Google Scholar
- Wang A., Tao W., Tong J., Gao J., Wang J., Hou G., ..., Zhu S. (2022) **m6A modifications regulate intestinal immunity and rotavirus infection** *eLife* **11** <https://doi.org/10.7554/eLife.73628>Google Scholar
- Wang J., Choi J.-M., Holehouse A. S., Lee H. O., Zhang X., Jahnel M., ..., Hyman A. A (2018) **A Molecular Grammar Governing the Driving Forces for Phase Separation of Prion-like RNA Binding Proteins** *Cell* **174**:688–699 <https://doi.org/10.1016/j.cell.2018.06.006>Google Scholar
- Wang J., Wang L., Diao J., Shi Y. G., Shi Y., Ma H., Shen H (2020) **Binding to m6A RNA promotes YTHDF2-mediated phase separation** *Protein Cell* **11**:304–307 <https://doi.org/10.1007/s13238-019-00660-2>Google Scholar
- Wang L., Hui H., Agrawal K., Kang Y., Li N., Tang R., ..., Rana T. M (2020) **m6A RNA methyltransferases METTL3/14 regulate immune responses to anti-PD-1 therapy** *EMBO J* **39**:e104514 <https://doi.org/10.15252/embj.2020104514>Google Scholar
- Wang L., Wen M., Cao X (2019) **Nuclear hnRNPA2B1 initiates and amplifies the innate immune response to DNA viruses** *Science* **365**:6454 <https://doi.org/10.1126/science.aav0758>Google Scholar
- Wang X., Lu Z., Gomez A., Hon G. C., Yue Y., Han D., ..., He C. (2014) **N6-methyladenosine-dependent regulation of messenger RNA stability** *Nature* **505**:117–120 <https://doi.org/10.1038/nature12730>Google Scholar
- Wang X., Zhao B. S., Roundtree I. A., Lu Z., Han D., Ma H., ..., He C. (2015) **N(6)-methyladenosine Modulates Messenger RNA Translation Efficiency** *Cell* **161**:1388–1399 <https://doi.org/10.1016/j.cell.2015.05.014>Google Scholar
- Wen Z, Zhong Z, Jr. D. J. (1995) **Maximal activation of transcription by Stat1 and Stat3 requires both tyrosine and serine phosphorylation** *Cell* **82**:241–250 [https://doi.org/10.1016/0092-8674\(95\)90311-9](https://doi.org/10.1016/0092-8674(95)90311-9)Google Scholar
- Winkler R., Gillis E., Lasman L., Safra M., Geula S., Soyris C., ..., Stern-Ginossar N. (2019) **m6A modification controls the innate immune response to infection by targeting type I interferons** *Nat Immunol* **20**:173–182 <https://doi.org/10.1038/s41590-018-0275-z>Google Scholar
- Xu W., He C., Kaye E. G., Li J., Mu M., Nelson G. M., ..., Shen H. (2022) **Dynamic control of chromatin-associated m(6)A methylation regulates nascent RNA synthesis** *Mol Cell* **82**:1156–1168 <https://doi.org/10.1016/j.molcel.2022.02.006>Google Scholar
- Yang X., Triboulet R., Liu Q., Sendinc E., Gregory R. I (2022) **Exon junction complex shapes the m(6)A epitranscriptome** *Nat Commun* **13**:7904 <https://doi.org/10.1038/s41467-022-35643-1>Google Scholar

- Yu H., Zhao K., Zeng H., Li Z., Chen K., Zhang Z., ..., Wu Z. (2021) **N6-methyladenosine (m6A) methyltransferase WTAP accelerates the Warburg effect of gastric cancer through regulating HK2 stability** *Biomed Pharmacother* **133**:111075 <https://doi.org/10.1016/j.biopha.2020.111075>Google Scholar
- Zhang H., Shao S., Zeng Y., Wang X., Qin Y., Ren Q., ..., Sun Y. (2022) **Reversible phase separation of HSF1 is required for an acute transcriptional response during heat shock** *Nat Cell Biol* **24**:340–352 <https://doi.org/10.1038/s41556-022-00846-7>Google Scholar
- Zhang Z., Theler D., Kaminska K. H., Hiller M., de la Grange P., Pudimat R., ..., Stamm S. (2010) **The YTH domain is a novel RNA binding domain** *J Biol Chem* **285**:14701–14710 <https://doi.org/10.1074/jbc.M110.104711>Google Scholar
- Zheng G., Dahl J. A., Niu Y., Fedorcsak P., Huang C. M., Li C. J., ..., He C. (2013) **ALKBH5 is a mammalian RNA demethylase that impacts RNA metabolism and mouse fertility** *Mol Cell* **49**:18–29 <https://doi.org/10.1016/j.molcel.2012.10.015>Google Scholar
- Zhong X., Yu J., Frazier K., Weng X., Li Y., Cham C. M., ..., Leone V. (2018) **Circadian Clock Regulation of Hepatic Lipid Metabolism by Modulation of m6A mRNA Methylation** *Cell Rep* **25**:1816–1828 <https://doi.org/10.1016/j.celrep.2018.10.068>Google Scholar

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Reviewer #1 (Public review):

Summary:

This study puts forth the model that under IFN- β stimulation, liquid-phase WTAP coordinates with the transcription factor STAT1 to recruit MTC to the promoter region of interferon stimulated genes (ISGs), mediating the installation of m6A on newly synthesized ISG mRNAs. This model is supported by strong evidence that the phosphorylation state of WTAP, regulated by PPP4, is regulated by IFN- β stimulation, and that this results in interactions between WTAP, the m6A methyltransferase complex, and STAT1, a transcription factor that mediates activation of ISGs. This was demonstrated via a combination of microscopy, immunoprecipitations, m6A sequencing, and ChIP. These experiments converge on a set of experiments that nicely demonstrate that IFN- β stimulation increases the interaction between WTAP, METTL3, and STAT1, that this interaction is lost with knockdown of WTAP (even in the presence of IFN- β), and that this IFN- β stimulation also induces METTL3-ISG interactions.

Strengths:

The evidence for the IFN- β stimulated interaction between METTL3 and STAT1, mediated by WTAP, is quite strong. Removal of WTAP in this system seems to be sufficient to reduce these interactions and the concomitant m6A methylation of ISGs. The conclusion that the phosphorylation state of WTAP is important in this process is also quite well supported. The authors have now also provided substantial evidence that phase separation of WTAP upon interferon stimulation facilitates m6A-methylation of multiple interferon stimulated genes.

<https://doi.org/10.7554/eLife.100601.2.sa3>

Reviewer #2 (Public review):

In this study, Cai and colleagues investigate how one component of the m6A methyltransferase complex, the WTAP protein, responds to IFN β stimulation. They find that viral infection or IFN β stimulation induces the transition of WTAP from aggregates to liquid droplets through dephosphorylation by PPP4. This process affects the m6A modification levels of ISG mRNAs and modulates their stability. In addition, the WTAP droplets interact with the transcription factor STAT1 to recruit the methyltransferase complex to ISG promoters and enhance m6A modification during transcription. The investigation dives into a previously unexplored area of how viral infection or IFN β stimulation affects m6A modification on ISGs. The observation that WTAP undergoes a phase transition is significant in our understanding of the mechanisms underlying m6A's function in immunity. However, there are still key gaps that should be addressed to fully accept the model presented.

Major points:

(1) More detailed analyses on the effects of WTAPsgRNA on the m6A modification of ISGs:

- a. A comprehensive summary of the ISGs, including the percentage of ISGs that are m6A-modified,
- b. The distribution of m6A modification across the ISGs, and
- c. A comparison of the m6A modification distribution in ISGs with non-ISGs.

In addition, since the authors propose a novel mechanism where the interaction between phosphorylated STAT1 and WTAP direct the MTC to the promoter regions of ISGs to facilitate co-transcriptional m6A modification, it is critical to analyze whether the m6A modification distribution holds true in the data.

(2) Since a key part of the model includes the cytosol-localized STAT1 protein undergoing phosphorylation to translocate to the nucleus to mediate gene expression, the authors should focus on the interaction between phosphorylated STAT1 and WTAP in Figure 4, rather than the unphosphorylated STAT1. Only phosphorylated STAT1 localizes to the nucleus, so the presence of pSTAT1 in the immunoprecipitate is critical for establishing a functional link between STAT1 activation and its interaction with WTAP.

(3) The authors should include pSTAT1 ChIP-seq and WTAP ChIP-seq on IFN β -treated samples in Figure 5 to allow for a comprehensive and unbiased genomic analysis for comparing the overlaps of peaks from both ChIP-seq datasets. These results should further support for their hypothesis that WTAP interacts with pSTAT1 to enhance m6A modifications on ISGs.

Minor points:

(1) Since IFN β is primarily known for modulating biological processes through gene transcription, it would be informative if the authors discussed the mechanism of how IFN β would induce the interaction between WTAP and PPP4.

(2) The authors should include mCherry alone controls in Figure 1D to demonstrate that mCherry does not contribute to the phase separation of WTAP. Does mCherry have or lack a PLD?

(3) The authors should clarify the immunoprecipitation assays in the methods. For example, the labeling in Fig. 2A suggests that antibodies against WTAP and pan-p were used for two immunoprecipitations. Is that accurate?

(4) The authors should include overall m6A modification levels quantified of GFPsgRNA and WTAPsgRNA cells, either by mass spectrometry (preferably) or dot blot.

Comments on revisions:

The authors thoroughly addressed the aforementioned points during the review process.

Reviewer #3 (Public review):

Summary:

This study presents a valuable finding on the mechanism used by WTAP to modulate the IFN- β stimulation. It describes the phase transition of WTAP driven by IFN- β -induced dephosphorylation. The evidence supporting the claims of the authors is solid.

Strength:

The key finding is the revelation that WTAP undergoes phase separation during virus infection or IFN- β treatment. The authors conducted a series of precise experiments to uncover the mechanism behind WTAP phase separation and identified the regulatory role of 5 phosphorylation sites. They also succeeded in pinpointing the phosphatase involved.

<https://doi.org/10.7554/eLife.100601.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

This study puts forth the model that under IFN-B stimulation, liquid-phase WTAP coordinates with the transcription factor STAT1 to recruit MTC to the promoter region of interferon-stimulated genes (ISGs), mediating the installation of m⁶A on newly synthesized ISG mRNAs. This model is supported by strong evidence that the phosphorylation state of WTAP, regulated by PPP4, is regulated by IFN-B stimulation, and that this results in interactions between WTAP, the m⁶A methyltransferase complex, and STAT1, a transcription factor that mediates activation of ISGs. This was demonstrated via a combination of microscopy, immunoprecipitations, m⁶A sequencing, and ChIP. These experiments converge on a set of experiments that nicely demonstrate that IFN-B stimulation increases the interaction between WTAP, METTL3, and STAT1, that this interaction is lost with the knockdown of WTAP (even in the presence of IFN-B), and that this IFN-B stimulation also induces METTL3-ISG interactions.

Strengths:

The evidence for the IFN-B stimulated interaction between METTL3 and STAT1, mediated by WTAP, is quite strong. Removal of WTAP in this system seems to be sufficient to reduce these interactions and the concomitant m⁶A methylation of ISGs. The conclusion that the phosphorylation state of WTAP is important in this process is also quite well supported.

Weaknesses:

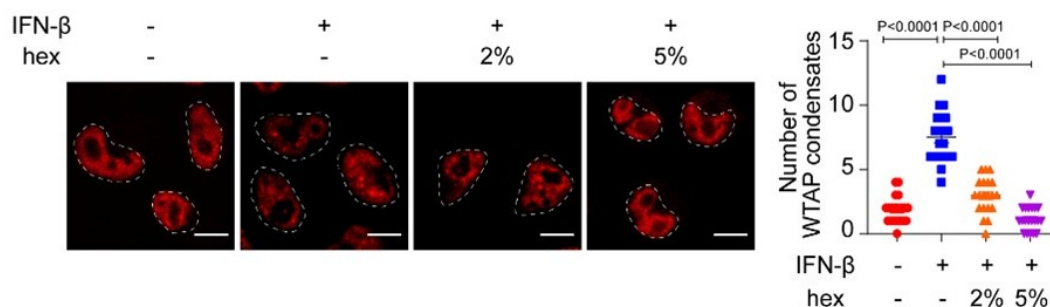
The evidence that the above mechanism is fundamentally driven by different phase-separated pools of WTAP (regulated by its phosphorylation state) is weaker. These experiments rely relatively heavily on the treatment of cells with 1,6-hexanediol, which has been shown to have some off-target effects on phosphatases and kinases (PMID 33814344).

Given that the model invoked in this study depends on the phosphorylation (or lack thereof) of WTAP, this is a particularly relevant concern.

We are grateful for the reviewer's positive comment and constructive feedback. 1,6-hexanediol (hex) was considered an inhibitor of hydrophobic interaction, thereby capable of dissolving protein phase separation condensates. Hex (5%-10% w/v) was still widely used to explore the phase separation characteristic and function on various protein, including some phosphorylated proteins such as pHSF1, or kinases such as NEMO1-3. Since hydrophobic interactions involved in various types of protein-protein interaction, the off-target effects of hex were inevitable. It has been reported that hex impaired RNA polymerase II CTD-specific phosphatase and kinase activity at 5% concentration⁴, which led to the reviewer's concern. In response to this concern, we investigated the phosphorylation level of WTAP under hex concentration gradient treatment. Surprisingly, we found that both 2% and 5% hex maintained the PPP4c-mediated dephosphorylation level of WTAP but still led to the dissolution of WTAP LLPS condensates (Figure 5-figure supplement 1H; Author response image 1), indicating that hex dispersed WTAP phase separation in a phosphorylation-independent manner. Consistent with our findings, Ge et al. used 10% hex to dissolve WTAP phase separation condensates⁵. Additionally, we found the phosphorylation level of STAT1 was not affected by both 2% and 5% hex, revealing the off-target and impairment function of hex on kinases or phosphatases might not be universal (Figure 5-figure supplement 1H). Collectively, since the 5% hex we used did not influence the de-phosphorylation event of WTAP, function of WTAP LLPS mediating interaction between methylation complex and STAT1 revealed by our results was independent from its phosphorylation status.

Author response image 1.

mCherry-WTAP-rescued HeLa cells were treated with 10 ng/mL IFN- β together with or without 2% or 5% hex and 20 μ g/mL digitonin for 1 hour or left untreated. Phase separation of mCherry-WTAP was observed through confocal microscopy. The number of WTAP condensates that diameter over 0.4 μ m of n = 20 cells were counted through ImageJ and shown. Scale bars indicated 10 μ m. All error bars, mean values $\pm$ SD, P-values were determined by unpaired two-tailed Student's t-test of n = 20 cells in (B). For (A), similar results were obtained for three independent biological experiments.



Related to this point, it is also interesting (and potentially concerning for the proposed model) that the initial region of WTAP that was predicted to be disordered is in fact not the region that the authors demonstrate is important for the different phase-separated states.

A considerable number of proteins undergo phase separation via interactions between intrinsically disordered regions (IDRs). IDR contains more charged and polar amino acids to present multiple weakly interacting elements, while lacking hydrophobic amino acids to

show flexible conformations⁶. In our study, we used PLAAC websites (<http://plaac.wi.mit.edu/>) to predict IDR domain of WTAP, and a fragment (234-249 amino acids) was predicted as prion-like domain (PLD). However, deletion of this fragment failed to abolish the phase separation properties of WTAP, which might be the main confusion to reviewers. To explain this issue, we checked the WTAP structure (within part of MTC complex) from protein data bank (<https://www.rcsb.org/structure/7VF2>) and found that the prediction of IDR has been renewed due to the update of different algorithm. IDR of WTAP expanded to 245-396 amino acids, encompassing the entire CTD region. Our results demonstrate that the CTD was critical for WTAP LLPS, as CTD deficiency significantly inhibited the formation of liquid condensates both *in vitro* and in cells. Also, phosphorylation sites on CTD were important for the phase transition of WTAP. These observations highlight the phosphorylation status on CTD region as a key driving force of phase separation, consistent with the established role of IDR in regulating LLPS. We have revised our description on WTAP IDR in our article following the reviewers' suggestion.

Taking all the data together, it is also not clear to me that one has to invoke phase separation in the proposed mechanism.

In this article, we observed that WTAP underwent phase transition during virus infection and IFN- β treatment, and proposed a novel mechanism whereby post translational modification (PTM)-driven WTAP LLPS was required for the m⁶A modification of ISG mRNAs. To verify the hypothesis, we first demonstrated the relationship between PTM and phase separation of WTAP. We constructed WTAP 5ST-D and 5ST-A mutant to mimic WTAP phosphorylation and dephosphorylation status respectively, and figured out that dephosphorylated WTAP underwent LLPS. We also found that PPP4 was the main phosphatase to regulate WTAP dephosphorylation. To further investigation, we introduced the potent PPP4 inhibitor, fostriecin. Consistent with our findings in PPP4 deficient model, fostriecin treatment significantly inhibited the IFN- β -induced dephosphorylation of WTAP and disrupted its LLPS condensates, indicating that PPP4 was the key phosphatase recruited by IFN- β to regulate WTAP, confirming that PTM governs WTAP LLPS dynamics (Figure 2-figure supplement 1C and H). Furthermore, fostriecin-induced impairment of WTAP phosphorylation and phase separation correlated with reduced m⁶A modification level and elevated ISGs expression level (Figure 4C and Figure 4-figure supplement 1E). These findings together emphasized that dephosphorylation is required for WTAP LLPS.

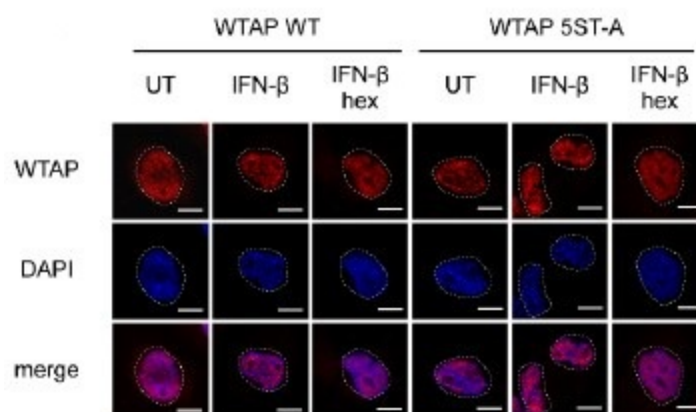
As for the function of WTAP LLPS, we assumed that WTAP might undergo LLPS to sequester STAT1 together with m⁶A methyltransferase complex (MTC) for mediating m⁶A deposition on ISG mRNAs co-transcriptionally under IFN- β stimulation. Given that hex dissolved WTAP LLPS condensates without affecting dephosphorylation status (Figure 5-figure supplement 1H; Author response image 1), various experiments we performed previously actually confirmed the critical role of WTAP LLPS during m⁶A modification (Figure 4A), as well as the mechanism that WTAP phase separation enhanced the interaction between MTC and STAT1 (Figure 5E-F). Subsequent to reviewer's comments, more experiments were conducted for further validation. We found the WTAP liquid condensates formed by wild type (WT) WTAP or WTAP 5ST-A mutant (which mimics dephosphorylated-WTAP) could be disassembled by hex, which also led to the impairment of the interaction between STAT1, METTL3 and WTAP (Figure 5-figure supplement 1E). In addition, *in vitro* experiments demonstrated that hex treatment significantly disrupted the interaction between recombinant GFP-STAT1 and mCherry-WTAP which expressed in the *E. coli* system (Figure 5F and Figure 5-figure supplement 1G). Notably, this prokaryotic expression system lacks endogenous phosphorylation machinery, resulting in non-phosphorylated mCherry-WTAP. For further validation, we performed the interaction of WTAP WT or 5ST-A with the promoter regions of ISG as well as the m⁶A modification of ISG mRNAs were attenuated by WTAP LLPS dissolution (Figure 4E and Figure 6E). These findings together revealed that WTAP LLPS were the critical

mediators of the STAT1-MTC interactions, ISG promoter regions binding and the co-transcription m⁶A modification.

Collectively, our results demonstrated that IFN- β treatment recruited PPP4c to dephosphorylate WTAP, thereby driving the formation of WTAP liquid condensates which were essential for facilitating STAT1-MTC interaction and m⁶A deposition, subsequently mediated ISG expression. Since we revealed a strong association between PTM-regulated WTAP phase transition and its m⁶A modification function, WTAP LLPS was a novel and functionally distinct mechanism that must be reckoned with in this study.

Author response image 2.

Wild type (WT) WTAP or 5ST-A mutant-rescued WTAP^{sgRNA} THP-1-derived macrophages are stimulated with or without with 10 ng/mL IFN- β together followed with 2% or 5% 1,6-hexanediol (hex) and 20 μ g/mL digitonin for 1 hour or left untreated. antibody and imaged using confocal microscope. Scale bars indicated 10 μ m.



Reviewer #2 (Public review):

In this study, Cai and colleagues investigate how one component of the m⁶A methyltransferase complex, the WTAP protein, responds to IFN β stimulation. They find that viral infection or IFN β stimulation induces the transition of WTAP from aggregates to liquid droplets through dephosphorylation by PPP4. This process affects the m⁶A modification levels of ISG mRNAs and modulates their stability. In addition, the WTAP droplets interact with the transcription factor STAT1 to recruit the methyltransferase complex to ISG promoters and enhance m⁶A modification during transcription. The investigation dives into a previously unexplored area of how viral infection or IFN β stimulation affects m⁶A modification on ISGs. The observation that WTAP undergoes a phase transition is significant in our understanding of the mechanisms underlying m⁶A's function in immunity. However, there are still key gaps that should be addressed to fully accept the model presented.

Major points:

- (1) More detailed analyses on the effects of WTAP sgRNA on the m⁶A modification of ISGs:*
 - a. A comprehensive summary of the ISGs, including the percentage of ISGs that are m⁶A-modified. merip-isg percentage*
 - b. The distribution of m⁶A modification across the ISGs. Topology*

c. A comparison of the m⁶A modification distribution in ISGs with non-ISGs. Topology

In addition, since the authors propose a novel mechanism where the interaction between phosphorylated STAT1 and WTAP directs the MTC to the promoter regions of ISGs to facilitate co-transcriptional m⁶A modification, it is critical to analyze whether the m⁶A modification distribution holds true in the data.

We appreciate the reviewer's summary of our manuscript and the constructive assessment. We have conducted the related analysis accordingly to present the m⁶A modification in ISGs in our model as reviewers suggested. Our results showed that about 64.29% of core ISGs were m⁶A modified under IFN- β stimulation (Figure 3-figure supplement 1B; Figure 3G), which was consistent with the similar percentage in previous studies[7,8]. The m⁶A distribution of the ISGs transcripts including *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was validated (Figure 3-figure supplement 1H).

Generally, m⁶A deposition preferentially located in the vicinity of stop codon, while m⁶A modification was highly dynamic under different cellular condition. However, we compared the topology of m⁶A deposition of ISGs with non-ISGs, and found that m⁶A modification of ISG mRNAs showed higher preference of coding sequences (CDS) localization compared to global m⁶A deposition (Figure 3H). Similarly, various researches uncovered the m⁶A deposition regulated by co-transcriptionally m⁶A modification preferred to locate in the CDS [9-11]. Since our results of m⁶A modification distribution of ISGs was consistent with the co-transcriptional m⁶A modification characteristics, we believed that our hypothesis and results were correlated and consistent.

(2) Since a key part of the model includes the cytosol-localized STAT1 protein undergoing phosphorylation to translocate to the nucleus to mediate gene expression, the authors should focus on the interaction between phosphorylated STAT1 and WTAP in Figure 4, rather than the unphosphorylated STAT1. Only phosphorylated STAT1 localizes to the nucleus, so the presence of pSTAT1 in the immunoprecipitate is critical for establishing a functional link between STAT1 activation and its interaction with WTAP.

Thank you for the constructive comments. As we showed in Figure 4, we found the enhanced interaction between phase-separated WTAP and the nuclear-translocated STAT1 which were phosphorylated under IFN- β stimulation, indicating the importance of phosphorylation of STAT1. We repeated the immunoprecipitation experiments to clarify the function of pSTAT1 in WTAP interaction. Our results showed that IFN- β stimulation induced the interaction of WTAP with both pSTAT1 and STAT1 (Figure 5-figure supplement 1C), indicating that most of the WTAP-associated STAT1 was phosphorylated. Taken together, our data proved that LLPS WTAP bound with nuclear-translocated pSTAT1 under IFN- β stimulation.

(3) The authors should include pSTAT1 ChIP-seq and WTAP ChIP-seq on IFN β -treated samples in Figure 5 to allow for a comprehensive and unbiased genomic analysis for comparing the overlaps of peaks from both ChIP-seq datasets. These results should further support their hypothesis that WTAP interacts with pSTAT1 to enhance m⁶A modifications on ISGs.

Thank you for raising this thoughtful comment. In this study, MeRIP-seq and RNA-seq along with immunoprecipitation and immunofluorescence experiments supported that phase transition of WTAP enhanced its interaction to pSTAT1, thus mediating ISGs m⁶A modification and expression by enhancing its interaction with nuclear-translocated pSTAT1 during virus infection and IFN- β stimulation. However, how WTAP-mediated m⁶A modification was related to STAT1-mediated transcription remained unclear. Several researches have revealed the recruitment of m⁶A methyltransferase complex (MTC) to transcription start sites (TSS) of

coding genes and R-loop structure by interacting with transcriptional factors STAT5B, SMAD2/3, DNA helicase DDX21, indicating the engagement of MTC mediated m⁶A modification on nascent transcripts at the very beginning of transcription [11-13]. These researches inspired us that phase-separated WTAP could be recruited to the ISG promoter regions via interacting with nuclear-translocated pSTAT1. To validate this mechanism, we have conducted ChIP-qPCR experiments targeting STAT1 and WTAP, revealed that IFN- β induced the comparable recruitment dynamics of both STAT1 and WTAP to ISG promoter regions (Figure 6A-B). Notably, STAT1 deficiency significantly abolished the bindings between WTAP and ISG promoter regions (Figure 6C). These findings established nuclear-translocated STAT1-dependent recruitment of phase separated WTAP and ISG promoter region, substantiated our hypothesis within the current study. We totally agree that ChIP-seq data will be more supportive to explore the mechanism in depth. We will continuously focus on the whole genome chromatin distribution of WTAP and explore more functional effect of transcriptional factor-dependent WTAP-promoter regions interaction in the future.

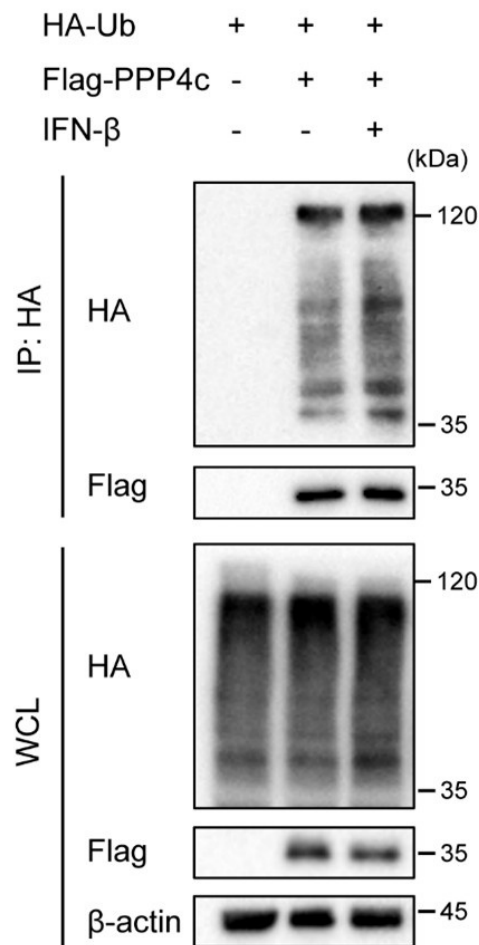
Minor points:

(1) Since IFN β is primarily known for modulating biological processes through gene transcription, it would be informative if the authors discussed the mechanism of how IFN β would induce the interaction between WTAP and PPP4.

Protein phosphatase 4 (PPP4) is a multi-subunit serine/threonine phosphatase complex that participates in diverse biologic process, including DDR, cell cycle progression, and apoptosis[14]. Several signaling pathway such as NF- κ B and mTOR signaling, can be regulated by PPP4. Previous research showed that deficiency of PPP4 enhanced IFN- β downstream signaling and ISGs expression, which was consistent with our findings that knockdown of PPP4C impaired WTAP-mediated m⁶A modification, enhanced the ISGs expression[15,16]. Since there was no significant enhancement in PPP4 expression level during 0-3 hours of IFN- β stimulation in our results, we explored the PTM that may influence the protein-protein interaction, such as ubiquitination. Intriguingly, we found the ubiquitination level of PPP4 was enhanced after IFN- β stimulation, which may affect the interaction between PPP4 and WTAP (Author response image 3). Further investigation between PPP4 and WTAP will be conducted in our future study.

Author response image 3.

HEK 293T transfected with HA-ubiquitin (HA-Ub) and Flag-PPP4 were treated with 10 ng/mL IFN- β or left untreated. Whole cell lysate (WCL) was collected and immunoprecipitation (IP) experiment using anti-Flag antibody was performed, followed with immunoblot. Similar results were obtained for three independent biological experiments.



(2) The authors should include mCherry alone controls in Figure 1D to demonstrate that mCherry does not contribute to the phase separation of WTAP. Does mCherry have or lack a PLD?

According to the crystal structure of mCherry protein in protein structure database RCSB-PDB, mCherry protein presents as a β -barrel protein, with no IDRs or multivalent interaction domains including PLD, indicating that mCherry protein has no capability to undergo phase separation. This characteristic makes it a suitable protein to tag and trace the localization or expression levels of proteins, and a negative control for protein phase separation studies. As the reviewer suggested, we include mCherry alone controls in the revised version (Figure 1D).

(3) The authors should clarify the immunoprecipitation assays in the methods. For example, the labeling in Figure 2A suggests that antibodies against WTAP and pan-p were used for two immunoprecipitations. Is that accurate?

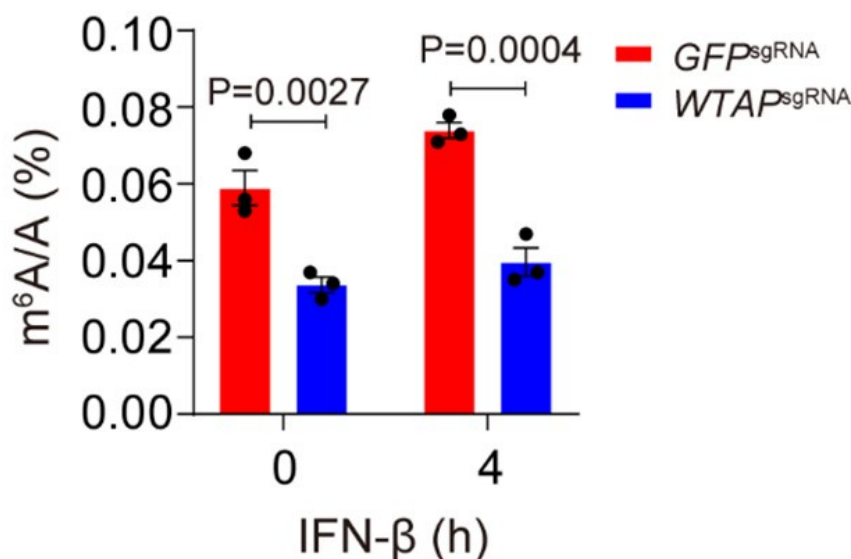
Due to the lack of phosphorylated-WTAP antibody, the detection of phosphorylation of WTAP was conducted by WTAP-antibody-mediated immunoprecipitation. We conducted immunoprecipitation to pull-down WTAP protein by WTAP antibody, then used antibody against phosphoserine/threonine/tyrosine (pan-p) to detect the phosphorylation level of WTAP. To avoid the misunderstanding, we have modified the description from pan-p to pWTAP (pan-p) in figures and revised the figure legends.

(4) The authors should include overall m^6A modification levels quantified of GFP^{sgRNA} and $WTAP^{sgRNA}$ cells, either by mass spectrometry (preferably) or dot blot.

We thank reviewer for raising these useful suggestions. As we showed in Figure 3F and J-K, the m^6A modification event and degradation of ISG mRNAs were significantly depleted in $WTAP^{sgRNA}$ cell lines, indicating that function of WTAP was deficient. Thus, we used the $WTAP^{sgRNA}$ #2 cell line to perform most of our experiment. Furthermore, we also found 46.4% of global m^6A modification was decreased in $WTAP^{sgRNA}$ THP-1 cells than that of control cells with or without IFN- β stimulation (Author response image 4), further validating that level of m^6A modification was significantly affected in $WTAP^{sgRNA}$ cells. Taken together, our data confirmed that m^6A methyltransferase activity was efficiently inhibited in our $WTAP^{sgRNA}$ cells.

Author response image 4.

Control (GFP^{sgRNA}) and $WTAP^{sgRNA}$ #2 THP-1-derived macrophages were treated with 10 ng/mL IFN- β for 4 hours. Global m^6A level was detected and quantified through ELISA assays. All error bars, mean values $\pm$ SEM, P-values were determined by two-way ANOVA test independent biological experiments.



Reviewer #3 (Public review):

Summary:

This study presents a valuable finding on the mechanism used by WTAP to modulate the IFN- β stimulation. It describes the phase transition of WTAP driven by IFN- β -induced dephosphorylation. The evidence supporting the claims of the authors is solid, although major analysis and controls would strengthen the impact of the findings. Additionally, more attention to the figure design and to the text would help the reader to understand the major findings.

Strength:

The key finding is the revelation that WTAP undergoes phase separation during virus infection or IFN- β treatment. The authors conducted a series of precise experiments to

uncover the mechanism behind WTAP phase separation and identified the regulatory role of 5 phosphorylation sites. They also succeeded in pinpointing the phosphatase involved.

Weaknesses:

However, as the authors acknowledge, it is already widely known in the field that IFN and viral infection regulate m⁶A mRNAs and ISGs. Therefore, a more detailed discussion could help the reader interpret the obtained findings in light of previous research.

We are grateful for the positive comments and the unbiased advice by the reviewer. To interpret our findings in previous research, we have revised the manuscript carefully and added more detailed discussion on ISG mRNAs m⁶A modification during virus infection or IFN stimulation.

It is well-known that protein concentration drives phase separation events. Similarly, previous studies and some of the figures presented by the authors show an increase in WTAP expression upon IFN treatment. The authors do not discuss the contribution of WTAP expression levels to the phase separation event observed upon IFN treatment. Similarly, METTL3 and METTL14, as well as other proteins of the MTC are upregulated upon IFN treatment. How does the MTC protein concentration contribute to the observed phase separation event?

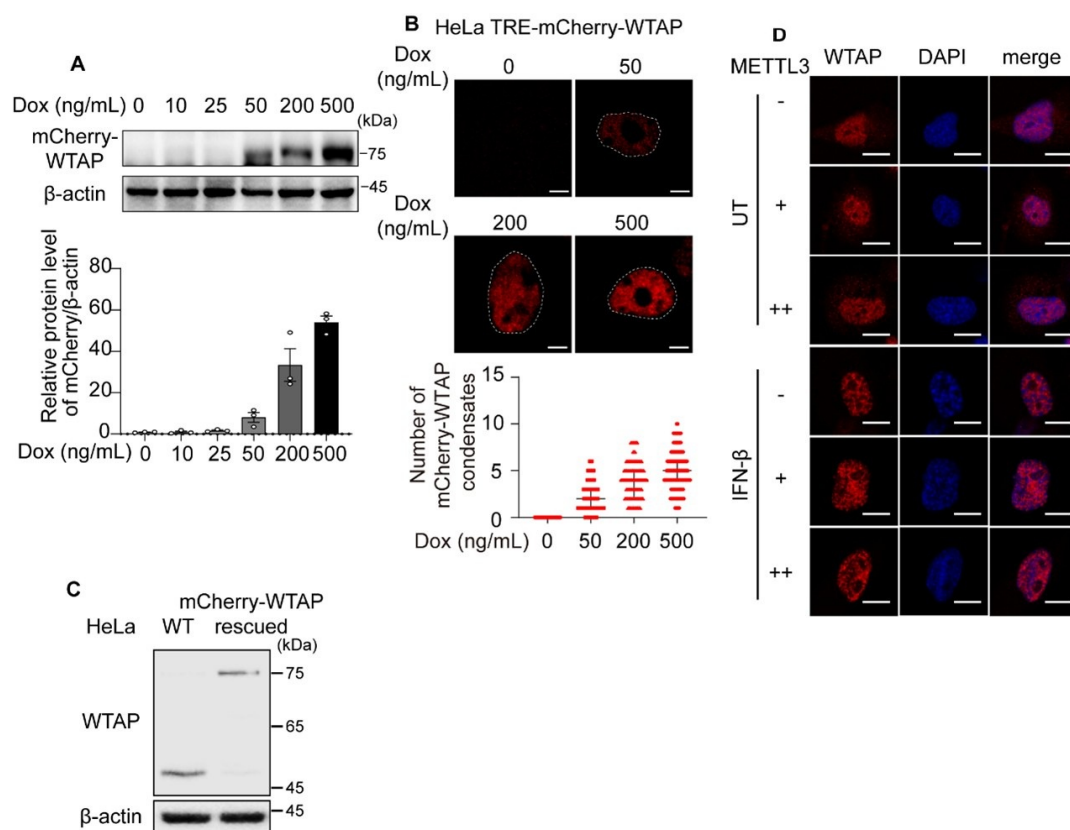
We thank reviewer for pointing out the importance of the concentration dependent phase transition. Previously, Ge et al. discovered that expression level of WTAP was up-regulated during LPS stimulation, thereby promoting WTAP phase separation ability and m⁶A modification, indicating that WTAP concentration indeed affects the phase separation event. In our article, we have generated the phase diagram with different concentration of recombinant mCherry-WTAP *in vitro* (Figure 1-figure supplement 1C). For in cells experiments, we constructed the TRE-mCherry-WTAP HeLa cells in which the expression of mCherry-WTAP was induced by doxycycline (Dox) in a dose-dependent manner (Author response image 5A). We detected the LLPS of mCherry-WTAP at different concentrations by increasing the doses of Dox, and found that WTAP automatically underwent LLPS only in an artificially high expression level (Author response image 5B). However, the cells we used to detect the WTAP phase separation in our article was mCherry-WTAP-rescued HeLa cells, in which mCherry-WTAP was introduced in WTAP^{sgRNA} HeLa cells to stably express mCherry-WTAP. We had adjusted and verified the mCherry-WTAP expression level precisely to make the protein abundance similar to endogenous WTAP in wild type (WT) HeLa cells (Author response image 5C). We also repeated the IFN-β stimulation experiments and found no significant increase of WTAP protein level (Figure 5-figure supplement 1A). These findings indicated the phase separation of WTAP in our article was not artificially induced due to the extremely high protein expression level.

MTC protein expression level was crucial for m⁶A modification during virus infection event. Rubio et al. and Winkler et al. revealed that WTAP, METTL3 and METTL14 were up-regulated after 24 hours of HCMV infection[8,17]. Recently, Ge et al. proposed that WTAP protein was degraded after 12 hours of VSV and HSV stimulation[5,18]. However, these studies only focused on the virus infection event, how the MTC protein expression change after direct IFN-β stimulation was still unclear. Our study investigated the transition change of WTAP under IFNβ stimulation in a short time, we detected the expression level of MTC proteins within 6 hours of IFN-β treatment, and found no significant enhancement of WTAP, METTL3 or METTL14 protein level and mRNA level (Figure 5-figure supplement 1B and Figure 5-figure supplement 1A;). Our *in vitro* experiments showed that introducing CFP-METTL3 protein have no significant influence on WTAP phase separation (Figure 4H). Additionally, we performed in cells experiments and found that increased expression of METTL3 had no effect on WTAP phase separation event (Author response image 5D). Taken together, WTAP phase

separation can be promoted by dramatically increased concentration of WTAP both *in vitro* and in cells, but the phase separation of WTAP under IFN- β stimulation in our study was not related with the expression level of MTC proteins.

Author response image 5.

(A) Immunoblot analysis of the expression of mCherry-WTAP in TRE-mCherry-WTAP HeLa cells treated with different doses of doxycycline (Dox). Protein level of mCherry-WTAP was quantified and normalized to β -actin of $n=3$ independent biological experiments. Results were obtained for three independent biological experiments. (B) Phase separation diagram of mCherry-WTAP in TRE-mCherry-WTAP HeLa cells treated with different doses of Dox were observed through confocal microscopy. Representative images for three independent biological experiments were shown in b while number of WTAP condensates that diameter over $0.4\ \mu\text{m}$ of $n=80$ cells were counted and shown as medium with interquartile range. Dotted white lines indicated the location of nucleus. Scale bars indicated $10\ \mu\text{m}$. (C) Immunoblot analysis of the expression of endogenous WTAP in wildtype (WT) HeLa cells and mCherry-WTAP-rescued $WTAP^{\text{sgRNA}}$ HeLa cells. (D) mCherry-WTAP-rescued HeLa cells were transfected with 0, 200 or 400 ng of Flag-METTL3, followed with 10 ng/mL IFN- β for 1 hour or left untreated (UT). Phase separation of mCherry-WTAP was observed through confocal microscopy. The number of WTAP condensates that diameter over $0.4\ \mu\text{m}$ of $n=20$ cells were counted through ImageJ and shown. Representative images of $n=20$ cells were shown. All error bars, mean values $\pm$ SD were determined by unpaired two-tailed Student's t-test of $n=3$ independent biological experiments in (A). For (A, C), similar results were obtained for three independent biological experiments.



How is PP4 related to the IFN signaling cascade?

Both reviewer #2 and reviewer #3 raised a similar point on the relationship between PPP4 and IFN signaling. In short, protein phosphatase 4 (PPP4) participates in diverse biologic process, including DDR, cell cycle progression and apoptosis¹⁴ and several signaling pathway. Previous research showed that deficiency of PPP4 enhanced IFN- β downstream signaling and ISGs expression, which was consistent with our findings that knockdown of PPP4C impaired WTAP-mediated m⁶A modification, and enhanced the ISGs expression^[15,16]. Since there was no significant enhancement in PPP4C expression level during 0-3 hours of IFN- β stimulation in our results, we tried to explore the post-translation modification which may influence the protein-protein interaction, such as ubiquitination. Intriguingly, we found the ubiquitination level of PPP4 was enhanced after IFN- β stimulation, which may affect the interaction between PPP4 and WTAP (Author response image 4). Investigation between PPP4 and WTAP will be conducted in our further study (also see minor points 1 of reviewer#2).

In general, it is very confusing to talk about WTAP KO as WTAPgRNA.

As we describe above, all WTAP-deficient THP-1 cells were generated using the CRISPR-Cas9 system with WTAP-specific sgRNA, and used bulk cells instead of the monoclonal knockout cell for further experiments. Since monoclonal knockout cells were not obtained, we refer it as WTAP^{sgRNA} THP-1 cells rather than WTAP-KO THP-1 cells. We confirmed that WTAP expression was efficiently knocked down in WTAP^{sgRNA} THP-1 cells, and the m⁶A modification level was significantly impaired (Figure 3I, Figure 3-figure supplement 1A and Author response image 4), which was suitable for mechanism investigation.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Related to the points raised in 'weaknesses' above, if the authors want to claim that this mechanism is reliant on WTAP phase-separated states, additional controls should be done to demonstrate this. Based on the available data it seems that it is just as likely that the phosphorylation state of WTAP is mediating interactions with other factors and/or altering its subcellular localization, which may or may not be related to phase separation.

We are grateful for the constructive suggestions. As we showed in , Figure 5-figure supplement 1H; Author response image 1 along with the explanation above, 5% hex dispersed the phase separation condensates of WTAP without affecting its phosphorylation status, proving the interaction between STAT1 and methylation complex impaired by hex was depended on WTAP LLPS but not its phosphorylation status (Figure 5E-H). To further confirmed the recruitment of WTAP LLPS on ISG promoter region, we performed the immunoprecipitation and ChIP-qPCR experiments of wild type (WT) WTAP, 5ST-D and 5ST-A rescued THP-1 cells. Our results uncovered the interaction between de-phosphorylated-mimic WTAP mutant and STAT1, and its binding ability with ISG promoter regions were depleted by hex without affecting its phosphorylation status (Author response image 2, Figure 5-figure supplement 1 F, Figure 6E). Taken together, we identified the de-phosphorylation event that regulated phase transition of WTAP during IFN- β stimulation, and proposed that LLPS of WTAP mediated by dephosphorylation was the key mechanism to bind with STAT1 and mediate the m⁶A modification on ISG mRNAs.

Reviewer #2 (Recommendations for the authors):

The author order is different for the main text and the supplementary file.

Thank you for pointing out this mistake. We have corrected it in our revised version.

Reviewer #3 (Recommendations for the authors):

Signaling molecules? Do you mean RNA or protein post-translational modification?

Thank you for pointing out this problem. In this sentence, we mean the post-translational modification of signaling proteins. We have corrected this mistake in our revised version.

Line 145: Do you mean Figure 1C?

We have corrected it in our revised version.

Line 214: Are the cells KO for WTAP? Do you mean CRISPR KO? In general, it is easier to present WTAPgRNA as WTAPKO.

Thank you for the constructive suggestion. As we explained above, in this project, all *WTAP*-deficient THP-1 cells were generated using the CRISPR-Cas9 system with *WTAP*-specific sgRNA, and used bulk cells instead of the monoclonal knockout cells. We confirmed that *WTAP* expression was efficiently knocked down in *WTAP*^{sgRNA} THP-1 cells, and the m⁶A modification level was significantly impaired (Figure 3I, Figure 3-figure supplement 1A and Author response image 4). Since monoclonal knockout cells were not obtained, we refer it as *WTAP*^{sgRNA} THP-1 cells rather than *WTAP*-KO THP-1 cells.

Line 221: WTAP KO has no effect on the expression level of transcription factors?

Thank you for pointing out the uncritical expression. We have corrected this in our revised version.

Figure 3C: I would suggest removing the tracks and showing the expression levels in TPMs.

According to the reviewer's suggestion, we have analyzed the results and showed the ISGs expression levels in fold change of TPMs (Figure 3D).

Figure 4C: It seems that the IP efficiency from METTL3 is lower in WTAP KO cells. That may impact the author's conclusions.

We have repeated the immunoprecipitation experiments of METTL3 and confirmed the immunoprecipitation (IP) efficiency from METTL3 had no difference between *WTAP*^{sgRNA} cells and the control cells (Figure 5C).

I would suggest performing an IP of WTAP with the 5StoA mutation to confirm the missing interaction with WTAP.

According to the reviewer's suggestion, we investigated the interaction between STAT1 and WTAP in WT cells and WTAP 5ST-A-rescued THP-1 cells. Our results showed that interaction between STAT1, METTL3 and WTAP were enhanced with WTAP 5ST-A mutation, which was depleted by hex treatment (Figure 5-figure supplement 1E). Thus, the interaction of WTAP WT or 5ST-A with the promoter regions of ISG were attenuated by WTAP LLPS dissolution (Figure 6E). Taken together, the interaction between STAT1 and MTC were relied on LLPS of WTAP.

In the graphical abstract, it is confusing to represent WTAP as a line. All other proteins are presented as circles. It is easy to confuse WTAP protein with an RNA. Additionally, m⁶A is too big in size. It should be smaller than a protein.

We thank the reviewer for raising this suggestion. We have modified the graphical abstract to avoid the confusion in our revised version (Figure 6F).

References

- (1) Wegmann, S., Eftekharzadeh, B., Tepper, K., Zoltowska, K.M., Bennett, R.E., Dujardin, S., Laskowski, P.R., MacKenzie, D., Kamath, T., Commins, C., et al. (2018). Tau protein liquid-liquid phase separation can initiate tau aggregation. *The EMBO journal* 37. 10.15252/emboj.201798049.
- (2) Lu, Y., Wu, T., Gutman, O., Lu, H., Zhou, Q., Henis, Y.I., and Luo, K. (2020). Phase separation of TAZ compartmentalizes the transcription machinery to promote gene expression. *Nat Cell Biol* 22, 453-464. 10.1038/s41556-020-0485-0.
- (3) Zhang, H., Shao, S., Zeng, Y., Wang, X., Qin, Y., Ren, Q., Xiang, S., Wang, Y., Xiao, J., and Sun, Y. (2022). Reversible phase separation of HSF1 is required for an acute transcriptional response during heat shock. *Nat Cell Biol* 24, 340-352. 10.1038/s41556-022-00846-7.
- (4) Duster, R., Kaltheuner, I.H., Schmitz, M., and Geyer, M. (2021). 1,6-Hexanediol, commonly used to dissolve liquid-liquid phase separated condensates, directly impairs kinase and phosphatase activities. *J Biol Chem* 296, 100260. 10.1016/j.jbc.2021.100260.
- (5) Ge, Y., Chen, R., Ling, T., Liu, B., Huang, J., Cheng, Y., Lin, Y., Chen, H., Xie, X., Xia, G., et al. (2024). Elevated WTAP promotes hyperinflammation by increasing m⁶A modification in inflammatory disease models. *J Clin Invest* 134. 10.1172/JCI177932.
- (6) Hou, S., Hu, J., Yu, Z., Li, D., Liu, C., and Zhang, Y. (2024). Machine learning predictor PSPire screens for phase-separating proteins lacking intrinsically disordered regions. *Nat Commun* 15, 2147. 10.1038/s41467-024-46445-y.
- (7) McFadden, M.J., McIntyre, A.B.R., Mourelatos, H., Abell, N.S., Gokhale, N.S., Ipas, H., Xhemalce, B., Mason, C.E., and Horner, S.M. (2021). Post-transcriptional regulation of antiviral gene expression by N6-methyladenosine. *Cell Rep* 34, 108798. 10.1016/j.celrep.2021.108798.
- (8) Winkler, R., Gillis, E., Lasman, L., Safra, M., Geula, S., Soyris, C., Nachshon, A., Tai-Schmiedel, J., Friedman, N., Le-Trilling, V.T.K., et al. (2019). m(6)A modification controls the innate immune response to infection by targeting type I interferons. *Nat Immunol* 20, 173-182. 10.1038/s41590-018-0275-z.
- (9) Li, Y., Xia, L., Tan, K., Ye, X., Zuo, Z., Li, M., Xiao, R., Wang, Z., Liu, X., Deng, M., et al. (2020). N(6)-Methyladenosine co-transcriptionally directs the demethylation of histone H3K9me2. *Nat Genet* 52, 870-877. 10.1038/s41588-020-0677-3.
- (10) Huang, H., Weng, H., Zhou, K., Wu, T., Zhao, B.S., Sun, M., Chen, Z., Deng, X., Xiao, G., Auer, F., et al. (2019). Histone H3 trimethylation at lysine 36 guides m(6)A RNA modification co-transcriptionally. *Nature* 567, 414-419. 10.1038/s41586-019-1016-7.
- (11) Barbieri, I., Tzelepis, K., Pandolfini, L., Shi, J., Millan-Zambrano, G., Robson, S.C., Aspris, D., Migliori, V., Bannister, A.J., Han, N., et al. (2017). Promoter-bound METTL3 maintains myeloid leukaemia by m(6)A-dependent translation control. *Nature* 552, 126-131. 10.1038/nature24678.
- (12) Hao, J.D., Liu, Q.L., Liu, M.X., Yang, X., Wang, L.M., Su, S.Y., Xiao, W., Zhang, M.Q., Zhang, Y.C., Zhang, L., et al. (2024). DDX21 mediates co-transcriptional RNA m(6)A modification to promote transcription termination and genome stability. *Mol Cell* 84, 1711-1726 e1711. 10.1016/j.molcel.2024.03.006.

- (13) Bhattarai, P.Y., Kim, G., Lim, S.C., and Choi, H.S. (2024). METTL3-STAT5B interaction facilitates the co-transcriptional m(6)A modification of mRNA to promote breast tumorigenesis. *Cancer Lett* 603, 217215. 10.1016/j.canlet.2024.217215.
- (14) Dong, M.Z., Ouyang, Y.C., Gao, S.C., Ma, X.S., Hou, Y., Schatten, H., Wang, Z.B., and Sun, Q.Y. (2022). PPP4C facilitates homologous recombination DNA repair by dephosphorylating PLK1 during early embryo development. *Development* 149. 10.1242/dev.200351.
- (15) Zhan, Z., Cao, H., Xie, X., Yang, L., Zhang, P., Chen, Y., Fan, H., Liu, Z., and Liu, X. (2015). Phosphatase PP4 Negatively Regulates Type I IFN Production and Antiviral Innate Immunity by Dephosphorylating and Deactivating TBK1. *J Immunol* 195, 3849-3857. 10.4049/jimmunol.1403083.
- (16) Raja, R., Wu, C., Basso, E.Y., Rubino, T.E., Jr., Utagawa, E.C., Magtibay, P.M., Butler, K.A., and Curtis, M. (2022). PP4 inhibition sensitizes ovarian cancer to NK cell-mediated cytotoxicity via STAT1 activation and inflammatory signaling. *J Immunother Cancer* 10. 10.1136/jitc-2022-005026.
- (17) Rubio, R.M., Depledge, D.P., Bianco, C., Thompson, L., and Mohr, I. (2018). RNA m(6) A modification enzymes shape innate responses to DNA by regulating interferon beta. *Genes Dev* 32, 1472-1484. 10.1101/gad.319475.118.
- (18) Ge, Y., Ling, T., Wang, Y., Jia, X., Xie, X., Chen, R., Chen, S., Yuan, S., and Xu, A. (2021). Degradation of WTAP blocks antiviral responses by reducing the m(6) A levels of IRF3 and IFNAR1 mRNA. *EMBO Rep* 22, e52101. 10.15252/embr.202052101.

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